IMAGE PROCESSING-BASED ANALYSIS OF EXTERNAL AND INTERNAL EGG QUALITY

A Thesis Presented to the Faculty of College of Computer Studies,
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In Partial Fulfillment of the Requirement for the Degree
Bachelor of Science in Computer Science
Specialized in *Intelligent Systems*

JONAS EMMANUEL G. BIEN
KENNETH S. KATIGBAK
JULIANA L. ZALAMEDA

MARIBEL V. CARDONA

JUNE 2026

APPROVAL SHEET

The thesis entitled "**IMAGE PROCESSING-BASED ANALYSIS OF EXTERNAL AND INTERNAL EGG QUALITY**" prepared and submitted by **Jonas Emmanuel G. Bien, Kenneth S. Katigbak and Juliana L. Zalameda,** in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science Specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

MARIBEL V. CARDONA
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of ____.

GENE MARCK B. CATEDRILLA
Member

CAROLINA R. JOVAL
Member

MYRA V. PAICAGLINO
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science specialized in Intelligent Systems.

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGEMENT

The researchers would like to express their heartfelt gratitude to all the individuals and offices who generously contributed their time, effort, and support to make this study successful.

The researchers would like to extend their deepest appreciation to the College of Computer Studies and their institution, Laguna State Polytechnic University, Los Baños Campus, for providing the facilities and resources that made this research possible. The researchers also would like to express their appreciation to Dr. Loyd S. Echalar, Associate Dean of the College of Computer Studies, for his valuable insights and professional guidance that greatly enhanced the quality of the research.

The researchers are also thankful to their professors and thesis panel for their constructive feedback and thoughtful recommendations that helped them refine and improve their work. The researchers owe profound gratitude to the thesis adviser, Prof. Maribel V. Cardona, for her patience, expertise, and constant encouragement throughout every stage of this study. Her mentorship played a vital role in completing this paper.

The Researchers

DEDICATION

This work is dedicated to the people who have shaped my journey, and whose love, wisdom, and sacrifices have made this achievement possible.

To the beloved parents, Mrs. Alicia S. Katigbak and Mr. Gil V. Katigbak, for their unwavering belief, their steadfast support, and the values they instilled in him. Their wisdom, encouragement, and selflessness have served as his guiding light throughout every step of him academic and personal pursuits.

To Mrs. Salve Bien and Mr. Conrad Bandian, whose enduring love and sacrifices have provided him with strength and inspiration. Their constant presence has been a source of resilience, motivation, and hope in realizing his aspirations.

To Mrs. Janet Zalameda and Mr. Jessy Zalameda Jr., for their silent yet powerful support, unwavering encouragement, and meaningful sacrifices. Their wisdom and compassion continue to inspire her daily, reinforcing the foundation that enabled her dreams to flourish.

K. S. K

J. E. G. B

J. L. Z.

ABSTRACT

Traditional methods of duck egg quality assessment rely heavily on manual candling and visual inspection, a process that is labor-intensive, subjective, and prone to human error, leading to inconsistent quality grading and financial losses. To address these inefficiencies, this study developed an automated, non-destructive image processing system capable of analyzing both external shell integrity and internal quality features using the YOLOv8-Large Segmentation deep learning architecture. The research utilized a custom-engineered LED candling setup to acquire a dataset of 3,720 duck eggs from a poultry farm in Los Baños, Laguna, resulting in 11,550 unique annotations across classes such as cracks, hairline cracks, air bubbles, and yolks. A comparative experimental framework was established to evaluate two model configurations: a Baseline Model trained on high-resolution raw imagery and an Enhanced Model utilizing Real-ESRGAN super-resolution pre-processing. The results demonstrated that the Baseline Model achieved superior performance with an overall accuracy of 88.5%, a Mean Average Precision (mAP@0.5) of 0.824, a Precision of 0.98, and an F1-Score of 0.80. In contrast, the Real-ESRGAN enhancement introduced artificial artifacts that significantly degraded the detection of microscopic defects, reducing the precision for hairline cracks from 0.85 to 0.47. The study concludes that high-fidelity raw image acquisition combined with the YOLOv8-Large Segmentation architecture provides a robust,real-time solution for automated egg sorting, effectively replacing subjective manual methods.

Keywords: *Egg Quality, Image Processing, YOLOv8-Large,Deep Learning, Real Time Detection*

TABLE OF CONTENTS

Preliminaries	**Page**
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	iv
Abstract	v
Table of Contents	vi
List of Figures	viii
List of Tables	ix
Definition of Terms	x
CHAPTER	
1 INTRODUCTION	
Background of the Study	1
Research Objectives	4
Significance of the Study	6
Scope and Limitations	7
2 REVIEW OF RELATED LITERATURE AND STUDIES	
Related Literature	11
Related Studies	11
Synthesis and Relevance of the Study	15
Conceptual Framework	19
3 METHODOLOGY	
Research Design	21
Project Locale	22
Materials	23
Model Development	
Data Gathering	25
Standardized Image Acquisition	26
Annotation Protocol	26
Image Pre-processing	27
Data Splitting	27
Experimental Framework	28
YOLOv8-Large Segmentation Architecture	29
Model Training	
Model A (Baseline Model)	30
Model B (Enhanced Model)	30
Device Prototype	32
System Deployment	33
4 RESULTS AND DISCUSSION	
Baseline Model (YOLOv8-Large Segmentation-Model A)	35
Enhanced Model (YOLOv8-Large Segmentation-Model B)	37
Model Evaluation Result	41
Comparative Performance Analysis of the Baseline and Enhanced Models	44

Comparative Analysis of Confusion Matrices 47
Success in Correct Detections (True Positives) 48
Evaluation of Missed Defects (False Negatives) 49
Training Performance and Model Convergence 52
YOLOv8 Training and Detection Results (Baseline Model) 56
YOLOv8-Large Segmentation with REAL-ESRGAN 58
Training and Detection Results (Enhanced Model)

5 SUMMARY, CONCLUSIONS AND RECOMMENDATION
Summary 60
Conclusion 62
Recommendations 65

REFERENCES
APPENDICES
Appendix A. Letter of Approval
Appendix B. Certificate of Acceptance
Appendix C. Evaluation Process & Validation
Appendix D. Statistical Validation Results
Appendix E. Labels Correlogram Visualization (Baseline Model)
Appendix F. Labels Correlogram Visualization (Enhanced Model)
Appendix G. YOLOv8 Training Source Code
Appendix H. Yolov8 Training Source Code with Real-ESRGAN of Baseline Model and Enhanced Model
Appendix I. Realtime Detection Source Code of Baseline Model and Enhanced Model
Appendix J. Confusion Matrix and Evaluation Metrics (Baseline Model)
Appendix K. Confusion Matrix and Evaluation Metrics (Enhanced Model)
Appendix L. Real Time Detection Output Result (Baseline Model)
Appendix M. Real Time Detection Output Result (Enhanced Model)
Appendix N. Grammarly and Turnitin Report
Appendix O. Publishable Paper (IEEE Format)
Appendix P. Curriculum Vitae

LIST OF FIGURES

No.	Title	Page
1	Conceptual Framework of the Study	18
2	Device Prototype	32
3	Detection of Internal and External Features of Duck Eggs (Baseline Model)	36
4	Detection of Internal and External Features of Duck Eggs(Enhanced Model)	37
5	Integration of Deep Learning Model for Duck Egg Quality Analysis	38
6	Real-time Egg Quality Analysis System (Baseline Model)	39
7	Real-time Egg Quality Analysis System (Enhanced Model)	40
8	Intermediate detection visualization during YOLOv8-Large Segmentation model training	56
9	Final detection results on test data showing classification of egg quality categories	57
10	Intermediate detection visualization during YOLOv8-Large Segmentation model training	58
11	Final detection results on test data showing classification of egg quality categories	59

LIST OF TABLES

No.	Title	Page
1	Literature Matrix	16
2	Materials Specification	23
3	Data Splitting	28
4	Precision Recall Curve	42
5	Precision–Confidence, Recall–Confidence, and F1–Confidence Curves	44
6	Comparative Analysis of True Positives	47
7	Comparative Analysis of False Negatives	48
8	Dataset Instances (Baseline Model)	50
9	Dataset Instances (Enhanced Model)	51
10	Training and Validation Baseline Model	53
11	Training and Validation Enhanced Model	54

DEFINITION OF TERMS

This section presents the definition of the terms used in the study, both technical and operational. The technical definitions provide the common meaning of each term; the operational definitions explain how each term will be defined within the study of egg quality analysis using image processing.

Air Cell	The pocket of air found between the eggshell and the inner membranes of an egg; its size is a key indicator of the egg's freshness and age.
Augment	The process of increasing the dataset by applying image modifications such as resizing, flipping, and brightness adjustments to improve the model's learning performance.
Candling	A traditional, non-destructive inspection method that uses a bright light source to illuminate the inside of an egg, revealing internal features such as the air cell, yolk, and albumen.
Crack Detection	The process of identifying any external shell defects, including visible and hairline cracks, which may compromise the safety and quality of the egg.
Dataset	The collection of annotated egg images used to train, validate, and test the YOLOv8-Large Segmentation model.
Deep Learning	An advanced subset of artificial intelligence that uses complex neural networks to automatically learn and identify patterns in data, such as detecting and classifying egg defects from images.
Egg Quality Assessment	The systematic evaluation of both external shell condition and internal contents using duck eggs to ensure they meet safety, nutritional, and market standards.

External Quality	The visible characteristics of an egg's shell, including its cleanliness, shape, texture, and integrity, indicate its physical soundness and market grade.
F1-Score	The harmonic mean of precision and recall, providing a balanced measure of the model's accuracy.
Hairline Crack	A fine, thin shell fracture detected on the egg's surface, considered one of the external quality indicators in this study.
Image Processing	The method of enhancing and analyzing images through techniques such as re-sizing, filtering, and contrast adjustment to improve detection of egg quality features.
Internal Quality	The condition of an egg's internal components, specifically the air cell, yolk, and albumen, determines its freshness, nutritional value, and suitability for consumption.
Machine Vision	The application of cameras and computer-based image analysis to automatically inspect and evaluate product quality, commonly used in agricultural and industrial settings.
Object Detection	A computer vision technique that identifies and locates specific features or defects within images, such as cracks or surface flaws on eggshells.
Precision	A performance metric that measures the proportion of correctly identified detections out of all detections made by the model.
Python	A versatile programming language used to develop and implement the image-processing algorithms and deep learning models applied in this research.

Recall	A metric that measures the proportion of correctly identified features out of all actual features present in the data.
Segmentation	The image processing step that isolates or highlights specific egg regions such as cracks, yolk, or air bubbles to enable clearer feature detection.
Yolk	The internal yellow component of the egg, analyzed in this study to assess internal condition and uniformity.
YOLOv8-Large Segmentation Model	Refers to the deep learning model used for object detection and classification of egg features, capable of identifying internal and external characteristics in real-time.

CHAPTER I

INTRODUCTION

Egg quality analysis plays a vital role in Philippine agriculture, providing a livelihood for small farmers and helping sustain the food supply for many rural communities. Numerous consumers demand a continuous shift toward safer, higher-quality egg products, making the need for reliable quality analysis even more urgent, as outlined in the Philippine National Standard for Fresh Duck Eggs (BAFS, 2015). The evaluation of egg quality in most farms, whether through shell appearance or internally through the condition of the yolk and albumen, has traditionally been conducted in a manual approach characterized as labor-intensive, subjective, and prone to human error that results in inconsistent quality, financial losses, and product rejection (Pan et al., 2007; Chen et al., 2020). Addressing this issue meets the demands of producers and consumers, with the help of researchers who have applied modern solutions such as image processing and machine learning techniques to automate quality inspection and reduce waste in produce.

The rapid advancement of digital imaging technologies has opened new opportunities for improving agricultural inspection processes. Image processing techniques enable the extraction of visual features from digital images, allowing automated systems to analyze shape, texture, color, and structural characteristics with high precision. When combined with machine learning and deep learning algorithms, these techniques can support real-time, non-destructive inspection systems capable of detecting subtle defects that may not be easily identified through manual methods. In recent years, deep learning-based object detection models have demonstrated strong

performance in various agricultural applications, including fruit grading, crop disease detection, and livestock monitoring.

Among these models, the You Only Look Once (YOLO) family of algorithms has attracted attention for its real-time detection capabilities and high computational efficiency (Jocher et al., 2023; Jiang et al., 2024). The latest iteration, YOLOv8-Large Segmentation, offers improved accuracy and speed, making it suitable for deployment in practical agricultural environments. By learning complex visual patterns from labeled image datasets, YOLOv8-Large Segmentation can identify multiple object classes within a single image, enabling simultaneous detection of different egg quality indicators. Its application in egg quality assessment provides an opportunity to automate both external and internal defect detection in a unified framework.

In addition to object detection, image enhancement techniques play an important role in improving model performance, particularly for low-contrast or poorly illuminated images. Internal egg inspection using candling often produces images with uneven lighting and limited visibility of internal features. To address this challenge, super-resolution enhancement methods such as Real-ESRGAN can be integrated into the pre-processing stage. Real-ESRGAN enhances image resolution and clarity by reconstructing fine details (Wang et al., 2021), thereby improving the visibility of internal structures such as yolks and air cells. The inclusion of an image enhancement component supports more reliable feature extraction and detection, especially under varying lighting conditions.

The relevance of improving egg quality inspection extends beyond technological advancement and aligns with broader global and national

development goals. The United Nations (2015) Sustainable Development Goals emphasize the importance of food security, responsible production, and agricultural innovation. In particular, SDG 2 (Zero Hunger) highlights the need to ensure access to safe and nutritious food, while SDG 12 (Responsible Consumption and Production) promotes reducing food losses through improved production systems. These goals underscore the importance of adopting technologies that enhance efficiency, accuracy, and sustainability in food production.

The Philippine Development Plan (PDP) 2023–2028 identifies agricultural modernization as a key strategy for improving productivity and resilience in the agri-food sector (NEDA, 2023). The PDP emphasizes adopting innovative technologies to address challenges such as low productivity, weak quality standards, and high post-harvest losses. By supporting automation and data-driven decision-making, technological solutions such as image processing-based inspection systems align with national efforts to strengthen food safety, improve product quality, and enhance competitiveness in the agricultural sector.

The present study investigated an image processing-based system for analyzing external and internal duck egg quality using deep learning techniques. The proposed approach integrates image pre-processing, enhancement, and object detection to identify egg quality indicators in a non-destructive and automated manner. The study evaluates the system's performance in terms of detection accuracy, reliability, and real-time applicability, aiming to provide a consistent and objective alternative to manual inspection methods. Through this investigation, the research

contributes to the growing body of work on intelligent agricultural systems and supports the broader goal of modernizing quality assessment practices in the poultry industry.

Research Objectives

The primary goal of this research was to develop an automated, non-destructive poultry egg quality assessment system by integrating Super-Resolution (Real-ESRGAN) and the YOLOv8-Large Segmentation deep learning architecture. The system was designed to provide a comprehensive evaluation of both external and internal quality markers, reducing the industry's reliance on subjective manual inspection.

Specifically, this study aimed to achieve the following:

1. Development of an Image Acquisition and Pre-processing Pipeline

- Design and construct a specialized image acquisition setup utilizing a 12MP CMOS sensor with f/1.8 aperture and a high-intensity LED candling system to capture consistent trans-illuminated (internal) and reflective (external) images.

- Implement a pre-processing module using Real-ESRGAN (Enhanced Super-Resolution Generative Adversarial Networks) to upscale and reconstruct captured images. This objective focused on recovering high-frequency spatial details such as hairline cracks and subtle internal bubble boundaries that are often lost due to sensor noise or low-light conditions during the candling process.

2. Implementation of the YOLOv8-Large Segmentation Detection Model

- Train and optimize the YOLOv8-Large Segmentation architecture to localize and classify a multi-class dataset of duck egg characteristics.
- External Feature Extraction: Achieve the precise localization of structural defects, including shell fractures, impact cracks, and microscopic hairline cracks.
- Internal Feature Extraction: Analyze internal markers revealed through candling, specifically focusing on the presence of air bubbles, yolk positioning, and transparency/viscosity indicators.

3. Performance Evaluation and System Validation

- Quantitatively assess system performance through standard computer vision metrics, including Mean Average Precision (mAP), Accuracy, Precision, Recall, and the F1-Score.
- Assess the system's operational viability for industrial deployment by measuring Inference Speed (Latency), ensuring the model can handle the high-volume throughput required in commercial poultry sorting facilities.

4. Operational and Environmental Analysis

- Analyze the strengths and limitations of the developed system in relation to varying environmental conditions, such as ambient light interference and surface cleanliness of the duck eggs.
- Determine the influence of dataset quality and the Real-ESRGAN enhancement on the final detection accuracy of the YOLOv8-Large Segmentation model.

Significance of the Study

The findings of this study are significant to various stakeholders involved in poultry production, food quality assurance, and agricultural research.

For poultry farmers and egg producers, the proposed image-processing-based system provides an automated, non-destructive method for assessing egg quality. By reducing reliance on manual inspection, the system may improve consistency in quality evaluation, minimize human error, and reduce post-harvest losses from undetected defects.

For food processors and quality control personnel, the system offers a reliable tool for detecting both external and internal egg defects in real time. This can support better quality grading, enhance food safety assurance, and improve compliance with quality standards in egg processing and distribution.

For agricultural researchers and developers, this study contributes to the application of deep learning and image enhancement techniques in agricultural quality inspection. The integration of object detection and super-resolution enhancement provides a reference framework for future studies exploring intelligent inspection systems for agricultural products.

For policymakers and agricultural institutions, the study supports national initiatives to modernize agriculture and reduce inefficiencies in food production systems. The proposed approach aligns with strategies that promote innovation, automation, and sustainability in the agri-food sector.

For future studies, the results of this research serve as a foundation for further exploration of automated egg quality assessment systems. The methodology and findings may be extended to other poultry eggs and refined through improved datasets, advanced models, and enhanced deployment strategies.

Scope and Limitations

This study focused on the development and evaluation of an image-processing–based system for automated assessment of poultry egg quality using deep learning techniques, specifically the YOLOv8-Large Segmentation object-detection model. While the proposed framework is designed to apply to poultry eggs in general, including chicken, quail, and other commercially produced eggs, the experimental implementation and validation of the system were conducted exclusively using duck eggs.

The study covered both external and internal egg quality assessment, including the detection and classification of visible shell defects (e.g., cracks and hairline cracks) and internal features (e.g., air bubbles and yolk characteristics). The label “Bad” strictly refers to eggs that fail quality standards due to physical shell defects or internal spoilage detected through candling inspection. Although the model includes separate detection classes such as crack and airbubble, these are treated as feature-level outputs, not final quality grades. Eggs predicted under the crack class are automatically categorized as “Bad” in the final assessment, as cracks weaken the shell and significantly increase the risk of contamination. Additionally, eggs are labeled “Bad” if the candling method reveals abnormal internal conditions, such as a

dark-colored yolk under light, which may indicate spoilage or quality degradation. In contrast, eggs are labeled “Good” when the yolk appears healthy under candling and a normal air bubble (air cell) is present, indicating proper egg structure and typical development. In summary, cracked eggs are always labeled “Bad,” while air bubble detections are further evaluated alongside internal candling results before assigning the final “Bad” or “Good” quality grade, ensuring a clear, structured classification ranking.

The research scope included image acquisition, pre-processing, enhancement, segmentation, data augmentation, and deep learning–based detection and classification. Real-ESRGAN (Wang et al. 2021) was integrated as an image enhancement module to improve image resolution and visual clarity before feature detection, particularly for subtle defects that are difficult to identify using raw images alone. Model training, validation, and testing were performed using the YOLOv8-Large Segmentation architecture (Jocher et al., 2023). The system's performance was evaluated using standard metrics, including accuracy, precision, recall, F1-score, and mean Average Precision (mAP), following established protocols for agricultural object detection (Jiang et al., 2024). The system was assessed under controlled lighting conditions and was intended to demonstrate real-time detection capability within a laboratory-scale setup.

The study did not include chemical, microbiological, or sensory evaluation of eggs, nor did it address egg freshness, nutritional content, or shelf-life analysis. Additionally, the system was developed as a proof-of-

concept and was not deployed in a full-scale commercial or industrial poultry processing environment. Several limitations were identified during the study.

First, although the model architecture is applicable to various poultry egg types, the dataset consisted solely of duck eggs, which may limit the direct generalizability of the results to other poultry species without further retraining and validation.

Second, class imbalance within the dataset, particularly for rare defects such as hairline cracks, affected detection performance. Despite using data augmentation techniques, the model showed lower accuracy in identifying subtle defects than in more visually distinct classes.

Third, lighting variability and image acquisition conditions influenced both image enhancement and detection accuracy. While Real-ESRGAN improved image clarity, it did not fully compensate for illumination and shadow inconsistencies introduced during data capture.

Fourth, the system relied exclusively on visual features extracted from images. Defects not visible through candling or surface inspection, as well as internal quality attributes requiring chemical or physical measurements, were beyond the capabilities of the proposed approach.

Finally, the system was evaluated in a controlled environment and has not yet been tested under real-world industrial conditions, such as high-speed conveyor systems or varying environmental factors. As a result, further optimization, dataset expansion, and field testing are required before practical large-scale deployment.

CHAPTER II

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature, Studies, and Systems

Egg quality assessment was established as a crucial aspect of poultry farming, directly impacting consumer preference, market value, and production efficiency. Traditional methods of quality analysis relied on manual inspection, which was often time-consuming, subjective, and inconsistent. With the advancements in artificial intelligence and computer vision, image processing has emerged as a powerful tool for automating and enhancing the accuracy of egg quality evaluation.

The Biological and Optical Properties of the Duck Egg

To justify the need for advanced AI, one must understand the physics of the medium. According to Singh et al. (2023) duck eggs (*Anas platyrhynchos*) possess a significantly higher shell density and thickness (averaging 0.35mm to 0.42mm) compared to *Gallus gallus* (chicken) egg.

- Light Attenuation and the Lambert-Beer Law (Wyatt, 1978)**:** When light from an LED passes through an eggshell, some of the light becomes weaker. This happens because the shell absorbs and scatters the light. Thicker shells block more light than thinner ones. Duck egg shells contain more calcium, so they are harder and denser compared to other poultry eggs. Because of this, the light scatters more strongly inside the shell, which can create bright glowing spots or a “blooming” effect in the captured image.
- Shell Mottling vs. True Cracks**:** Duck eggs often exhibit "mottling," the translucent spots caused by moisture distribution in the shell. According

to Wong et al. (2020) a standard CNN, mottling can look identical to a fracture. The theoretical necessity of high-precision detection lies in differentiating between these harmless biological variations and actual structural failures through deep spatial analysis.

External and Internal Quality Assessment

External analysis focuses on the integrity and shape of the shell. Wong et al. (2020) established the mathematical classification of egg shapes, which influences physical stability. For defect detection, early systems like those by Pan et al. (2007) used geometric parameters and neural networks to achieve 96.8% accuracy in crack detection.

Internal quality assessment, traditionally done via manual candling, has also evolved. Hyperspectral imaging was successfully used by Zhang et al. (2015) to detect internal defects like scattered yolks and bubbles with high precision. Similarly, Nakaguchi and Ahamed (2023) utilized thermal imaging combined with deep learning to detect early-stage embryos. These studies confirm that non-invasive imaging is superior to manual checks for preserving egg integrity.

Evolution of AI in Egg Inspection: From SVM to CNN

The transition from traditional image processing to AI is well-documented. Early machine learning approaches utilized Support Vector Machines (SVM). Yang et al. (2014) and Chen et al. (2020) achieved accuracies ranging from 98% to 98.75% using SVMs for crack detection. However, these methods often required rigid feature extraction protocols.

Convolutional Neural Networks (CNNs) marked a significant leap forward. Ma et al. (2017) demonstrated that CNNs outperformed Fisher's

Linear Discriminant (FLD) in classifying double-yolk eggs, reducing processing time from 0.20s to 0.12s. This highlighted the efficiency of deep learning for real-time applications.

Advancements in YOLO and Object Detection

The "You Only Look Once" (YOLO) architecture revolutionized detection by framing it as a regression problem, enabling real-time processing. Recent studies have adapted YOLO for poultry. Jiang et al. (2024) proposed YOLOv8-PG, a lightweight model for pigeon eggs that reduced parameter count for mobile robots. Cheng et al. (2024) used a multi-domain training strategy with YOLOv8, achieving 88.8% mAP and proving its robustness against environmental variations. These adaptations underscore YOLO's expanding role as a practical backbone for high-speed, automated egg quality monitoring systems in modern poultry applications.

Technical Deep-Dive: Architecture of YOLOv8-Large

Many agricultural studies use small models because they run faster. However, this research used YOLOv8-Large Segmentation to focus on detecting very small and subtle defects, such as tiny cracks that are hard to see. YOLOv8-Large Segmentation also uses an anchor-free detection system, which means it predicts objects directly without relying on preset box guides, making detection more flexible and precise.

The YOLOv8-Large Segmentation architecture has three main parts:

- **Backbone (Feature Extractor)**

  This is the part that scans the egg image and identifies important visual details. The Large version has more internal layers than the smaller versions, helping the model understand complex light patterns and fine

shell details better. This allows it to capture even the smallest irregularities in the egg.

- **Neck (Feature Combiner)**

  This section connects and blends different levels of image information. It makes sure that small details (like the thin line of a crack) are combined with bigger features (like the yolk or overall egg shape) so nothing important is missed.

- **Head (Detection Output Layer)**

  In YOLOv8-Large Segmentation, identifying *what* the defect is (classification) and *where* it is (location) are processed separately. This improves accuracy, especially for detecting very thin cracks or hairline fractures.

Image Enhancement and Pre-processing (Real-ESRGAN)

Pre-processing is vital for improving detection in challenging conditions. Neamah et al. (2024) found that Adaptive Histogram Equalization (AHE) significantly improved crack detection accuracy. Furthermore, recent advancements in super-resolution, specifically Real-ESRGAN, have been applied to enhance low-quality inputs for object detection. Correa et al. (2025) and Deshpande (2025) demonstrated that using Real-ESRGAN to upsample thermal and low-light images significantly improves the feature extraction capabilities of downstream YOLO detectors, suggesting that enhancing image resolution before processing is a viable strategy for detecting subtle defects like hairline cracks. This also supports its feasibility for identifying fine-grained structural irregularities in eggshell surfaces. Additionally, applying super-resolution in the pre-processing stage strengthens model sensitivity to micro-

fractures that are often missed in raw low-resolution captures. Consequently, this approach improves the system's capacity to detect hairline defects while preserving the non-destructive nature of the assessment pipeline. By enhancing critical visual details prior to inference, the pre-processing pipeline plays a key role in enabling more reliable defect localization and segmentation for practical poultry egg quality analysis.

Theoretical Mechanics of Real-ESRGAN Enhancement

Agricultural images captured through candling can sometimes be blurry or noisy. Real-ESRGAN is used in this study to solve a common challenge: recovering clear, high-resolution details from low-quality images. This is known as an inverse problem; the model tries to rebuild details that are not clearly visible in the original capture. Real-ESRGAN improves images using two neural networks that compete and learn from each other:

1. **Generator (Image Enhancer)**

   This part takes a low-quality candling image of a duck egg and creates a sharper, 4x larger (upscaled) version. Since some details are missing or unclear in the original image, the Generator predicts and fills in the missing pixels to make the image look more detailed. It uses a special block called RRDB, which helps keep the real structure of the egg while reducing noise from the camera sensor.

2. **Discriminator (Image Checker)**

   This network checks the upscaled image and tries to tell whether it looks like a real high-resolution egg photo or an artificially enhanced one. As both networks improve, the cracks—especially very thin hairline cracks that may only appear as 1–2 pixels in the raw image—become more

visible and more contrasted. This makes them easier for the YOLOv8-Large model to detect and analyze accurately.

Synthesis and Relevance of the Study

The reviewed literature established a clear progression tracing a path from human-dependent processes to high-speed autonomous systems. This progression began with manual inspection, a labor-intensive approach that was often limited by subjectivity and human error. The field transitioned into automated territory with the adoption of Support Vector Machines (SVM), which introduced statistical learning and hand-crafted feature extraction to improve consistency. This was followed by a paradigm shift toward Convolutional Neural Networks (CNN), which revolutionized the field by enabling deep learning models to automatically learn complex spatial hierarchies of features. Most recently, the literature highlighted the emergence of the YOLO (You Only Look Once) framework as the most advanced approach, which optimized these neural architectures for real-time application by consolidating object detection into a single, high-speed pipeline.

While SVMs (Yang et al., 2014) and standard CNNs (Ma et al., 2017) offer high accuracy, they often lack the speed required for real-time industrial sorting or require complex, manual feature engineering. Lightweight models like YOLOv8-PG (Jiang et al., 2024) and HCES-YOLO (Zhimin et al., 2024) prioritize speed and low computational cost, primarily for mobile robotics. However, these lightweight architectures often sacrifice depth and parameter count, which can limit their ability to detect extremely fine details like hairline cracks. This creates an opportunity for larger segmentation models, such as

YOLOv8-Large, to balance real-time inference with improved sensitivity to micro-defects in poultry egg quality assessment.

Table 1. Literature Matrix

Technology Category	Core Research Example	mAP / Accuracy	Technical Strength	Primary Limitation
Traditional ML	Yang et al. (2014)	98.07%	High accuracy in controlled lab settings.	Manual feature extraction required.
Traditional CNN	Ma et al. (2017)	98.00%	Autonomous feature learning.	High latency for real-time sorting.
Lightweight YOLO	Jiang et al. (2024)	82.5%	Extremely fast for mobile devices.	Misses microscopic/ hairline cracks.
Enhanced YOLOv8L-seg	This Study	Target > 85%	High sensitivity via Real-ESRGAN.	High computational cost for training.

This study uses the YOLOv8-Large Segmentation model to solve the specific weaknesses found in earlier egg-quality detection methods. In the world of AI, there is usually a trade-off: you either get high speed or high accuracy. Smaller and "lightweight" YOLO models are built for speed, making them popular for mobile robots, but they have "shallow" networks—meaning they only look at the most obvious parts of an image. Because duck egg defects are often microscopic, this study choose the Large version of the model for a much deeper network. This allows it to analyze complex and tiny details that smaller models would simply overlook. This deeper structure is critical for detecting small defects. For example, a hairline crack on a duck

egg can be thinner than a human hair and easily confused with the natural grain of the shell. Similarly, during candling, the edges of the yolk or the size of the air cell can appear blurry. While smaller models might confuse these important details with image noise, the YOLOv8-Large Segmentation model is powerful enough to distinguish a structural crack from a harmless surface stain.

To further enhance model accuracy, this research integrates an image enhancement stage inspired by studies utilizing Real-ESRGAN, a deep learning-based super-resolution technique. Prior to classification, each egg image undergoes digital enhancement to reduce noise, refine texture details, and sharpen structural edges. This enhancement step also improves the visibility of subtle eggshell patterns and micro-defects that may be overlooked in raw images. Consequently, the refined images provide more discriminative features, enabling the classification model to achieve higher prediction reliability.This process functions similarly to a high-resolution pre-processing filter that amplifies edge contrast and improves visual clarity, allowing subtle shell irregularities—particularly hairline cracks—to become more distinguishable from the surrounding background. Combining the "strong brain" of YOLOv8-Large Segmentation with these high-resolution images, this study aims to achieve the best of both worlds: the extreme accuracy of traditional, heavy deep-learning models and the rapid, real-time speed that the YOLO framework is famous for. The result is a system that is not only fast enough to keep up with industrial production but also accurate enough to ensure that only high-quality eggs reach the consumer.

Conceptual Framework

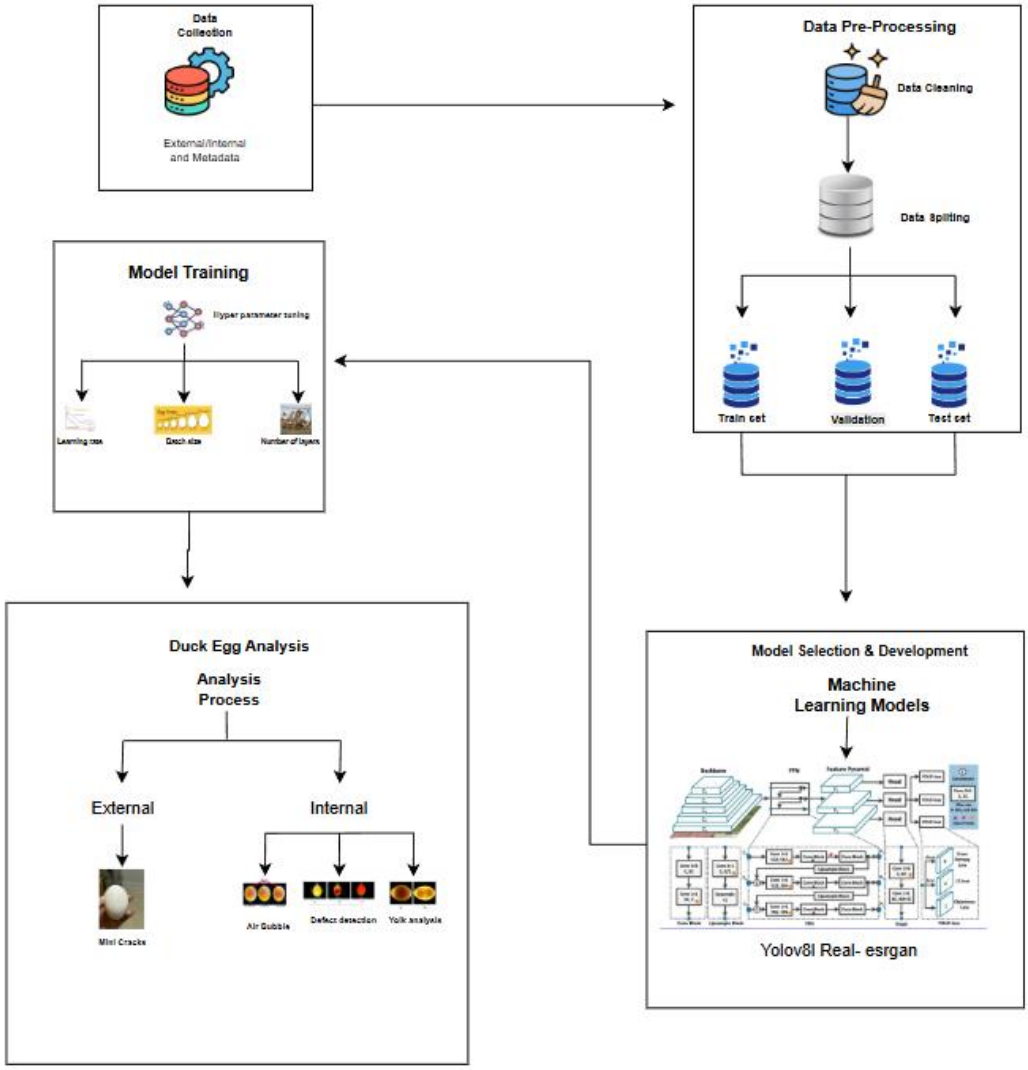


Figure 1. Conceptual Framework of the Study

The conceptual framework illustrates the complete workflow of the proposed egg quality assessment system using image processing and deep learning techniques. The process begins with data collection, where images of duck eggs are captured to represent both external and internal quality attributes. External data focus on shell surface conditions, particularly the presence of mini cracks and hairline fractures. Internal data include air bubble size, defect detection, and yolk analysis, which are obtained through the

candling method, a non-destructive technique that illuminates the egg to reveal its internal structure.

After data acquisition, the images undergo data pre-processing, which includes data cleaning to remove noisy or low-quality images and ensure consistency across the dataset. The cleaned dataset is then split into training, validation, and testing sets to support effective model learning and unbiased performance evaluation.

The model selection and development stage involves the use of machine learning and deep learning architectures. In this study, the YOLOv8-Large Segmentation model, enhanced with Real-ESRGAN, is employed to improve image quality and support accurate feature extraction. This model is designed to detect and localize external shell defects and assist in analyzing internal quality indicators from candling images.The model training phase includes hyperparameter tuning, where learning rate, batch size, and number of epochs are optimized to achieve a balance between accuracy and real-time performance. The trained models are evaluated using the validation and test datasets to ensure robustness and generalization.

The egg analysis process integrates both external and internal assessment results. The system performs real-time detection and classification of shell mini cracks for external analysis, while internal analysis evaluates air bubbles, internal defects, and yolk characteristics. This integrated framework enables a fast, accurate, and automated duck egg quality evaluation system, supporting efficient quality control in poultry production.

CHAPTER III

METHODOLOGY

This chapter outlines the research design and methodological framework established to develop a robust, technology-driven system for the comprehensive assessment of both the internal and external quality of duck eggs. To overcome the limitations of traditional manual inspection, which was historically labor-intensive and prone to human error, the study employed a systematic approach centered on non-destructive computer vision and deep learning architectures. The methodology also emphasizes reproducibility and scalability to ensure the system can be effectively implemented in diverse poultry farm environments.

The primary objective was to create a solution capable of high-speed, real-time evaluation without compromising accuracy. The framework covered the entire developmental pipeline, which began with standardized image acquisition through a custom-built candling setup to ensure consistent lighting. This was followed by the implementation of image enhancement techniques, such as Real-ESRGAN, and the comparative evaluation of different deep learning models to determine their effectiveness in a production-like environment.By analyzing the raw data and enhanced imagery, this methodology sought to identify the optimal configuration for detecting various defects ranging from subtle external hairline cracks to internal air cell and yolk irregularities. Ultimately, the structured approach was designed to fulfill the specific technical and operational goals outlined in the study's objectives,

ensuring that the resulting system remained both industrially viable and highly precise.

Research Design

This study utilizes a combined developmental and experimental applied research design to engineer and evaluate an automated diagnostic system for duck egg quality assessment. The developmental phase is dedicated to the systematic creation of a high-fidelity image corpus and a standardized acquisition environment. For this phase, a total of 3,720 physical duck eggs were procured and captured using a high-resolution 12MP CMOS sensor to ensure sufficient spatial detail for microscopic defect detection. To ensure scientific consistency and minimize stochastic noise, the images were captured inside a custom-engineered candling box with a 10W LED light source and a light-sealed chamber. By locking the camera at a fixed 25 cm focal distance and stabilizing the sensor's exposure parameters, we ensured that the model learned to identify actual quality features—such as yolk and shell rather than reacting to variations in lighting or perspective. The data engineering process further refined this dataset through polygon-based instance segmentation, resulting in a dense corpus of 11,550 unique annotations. This developmental effort culminated in the integration of a real-time detection pipeline using the OpenCV library, transforming the theoretical models into a functional tool capable of providing instantaneous "Good/Bad" quality feedback in a simulated production environment.

The experimental phase follows a comparative framework designed to test the impact of Generative Adversarial Networks (GANs) on computer

vision accuracy in agriculture. Specifically, this phase centers on a head-to-head evaluation between two distinct model configurations: Model A (Baseline) and Model B (Enhanced). Both models utilize the YOLOv8-Large Segmentation architecture, selected for its deep hierarchical feature extraction and high spatial granularity. Model A serves as the control group, trained on raw 12MP imagery to establish the performance ceiling of standard high-resolution data. Conversely, Model B represents the experimental group, where the raw images are first processed through Real-ESRGAN as an offline pre-processing step in Google Colab.

This experimental setup quantifies the 'performance lift'—or potential degradation—introduced by GAN-based super-resolution on agricultural imagery.The models are evaluated using standard performance metrics precision, recall, and F1-score to assess how effectively the system identifies and classifies quality indicators across the dataset. By combining system development with a comparison, the research design enables a clear assessment of whether super-resolution techniques are essential for achieving high-precision agricultural grading, or whether a well raw imaging setup can provide sufficient reliability for real time applications.

Project Locale

This study was conducted in Los Baños, Laguna, where data were collected from a selected poultry farm. The research site was chosen primarily due to its accessibility, which allowed for efficient data collection, and its steady availability of duck eggs that met the requirements of the study. The raw data obtained from this location consisted of various physical and quality features of the duck eggs. These features provided essential information

needed to examine the condition and characteristics of the samples. Both internal and external attributes of the eggs were considered in the data collection process, allowing for a more detailed assessment of egg quality. The information gathered from the poultry farm served as the primary basis for analyzing the internal and external characteristics of the duck eggs and supported the overall objectives of the research.

Materials

The relevant resources used in this research to develop and analyze the duck egg quality evaluation using modern image processing and deep learning techniques.

Table 2. Materials Specification

Materials	Description
Software Specifications	A set of software frameworks, programming tools, and cloud platforms were used to develop and optimize the YOLOv8-Large Segmentation model and the overall egg quality analysis system.
Python 3.10+	Python was the main language used in the project. All pre-processing scripts, training workflows, and real-time detection codes were written in Python. Its wide library support made it much easier to handle image operations, run models, and combine different parts of the system into one working program.
PyTorch	Since YOLOv8 runs on the PyTorch framework, this study used PyTorch for training and fine-tuning the segmentation models. It allowed the researchers to experiment quickly, adjust parameters as needed, and make use of GPU acceleration when training in Google Colab. PyTorch helped streamline both model development and comparison between the standard YOLOv8-Large Segmentation and the YOLOv8-Large Segmentation combined with Real-ESRGAN.
TensorFlow	TensorFlow was not the main training framework, but it was used during the early stages of the study to validate pre-processing outputs, test initial classification Baselines, and support comparison of results. Some

TensorFlow tools were also used for visualizing image features and checking consistency across datasets.

OpenCV

OpenCV was used heavily during pre-processing. It handled tasks like resizing, cleaning, and enhancing the candling images before they were uploaded to Roboflow or passed into the model. OpenCV was also responsible for managing the live camera feed during real-time testing, making it possible to display segmentation masks and predictions smoothly on-screen.

Roboflow

Roboflow served as the main platform for preparing the dataset. All collected images were uploaded to Roboflow, where the segmentation masks for cracks, yolk outlines, internal shadows, and other features were drawn manually. The platform also generated augmented versions of the dataset so the model could learn from more diverse samples. After annotation, the dataset was exported in YOLOv8 format and used directly for training in Google Colab.

Google Colab

Google Colab provided the GPU resources needed to train the models. The large size of the dataset and the complexity of YOLOv8-Large Segmentation made GPU training essential. Colab allowed the researchers to train both the standard model and the enhanced version with Real-ESRGAN without requiring high-end local hardware. Once training was completed, the final model was downloaded and tested offline.

Hardware Specifications

To collect consistent and high-quality images, the study used the following equipment:

12MP High-Resolution Camera

A 12-megapixel camera was used to capture both external and internal images of the duck eggs. The clarity provided by the camera was important, especially for identifying internal and external features during candling. Using the same camera throughout the project also helped maintain consistency between the training data and the real-time tests.

Custom-Built LED Candling Box

A custom LED candling box was designed and built specifically for this research. It provided a uniform lighting environment, which helped reveal internal egg features such as the yolk position, air cell size, and albumen clarity. The controlled light source minimized shadows and reflections, which made the internal images easier to annotate and improved the model’s accuracy during training.

Laptop	The real-time egg quality analysis system was implemented and tested on a laptop equipped with an 11th Generation Intel® Core™ i5-1135G7 processor system had 8 GB of RAM to support efficient execution of deep learning models and image processing tasks. A 500 GB solid-state drive (SSD) provided fast storage and data access for high-resolution images. The laptop used a 64-bit architecture, ensuring compatibility with the required software frameworks and libraries for real-time model deployment and evaluation.

Model Development

Data Gathering

The foundation of this diagnostic system is a specialized dataset designed to capture the natural variability of duck egg quality. A total of 3,720 duck eggs were collected from a commercial poultry farm in Laguna, spanning a wide range of conditions from "Grade A" quality to various internal and external defects.

A critical distinction in this study is the relationship between physical samples and dataset instances. While we photographed 3,720 unique eggs, the final dataset comprises 11,550 unique annotations. The reasoning behind this is that a single egg often presents multiple quality indicators simultaneously—for example, an egg may possess a displaced yolk while also exhibiting a micro-fracture on the shell. By labeling these as separate instances, we provided the model with a "multi-label" learning environment, which is essential for a truly standardized and comprehensive assessment

system. Every egg was manually inspected and categorized by poultry experts to ensure that the ground-truth labels were scientifically accurate before being used for supervised learning.

Standardized Image Acquisition (iPhone XR Setup)

High-resolution imagery was captured using the built-in 12MP camera of an iPhone XR (Apple Inc.). The device features a 1/2.55" CMOS sensor with a fixed physical aperture of f/1.8. To ensure dataset uniformity and eliminate the variability typically associated with smartphone photography, a strictly controlled acquisition environment was constructed:

- External Analysis: The camera was positioned at a fixed focal distance of 25 cm above the samples on a matte-black, non-reflective surface to highlight shell fractures and eliminate background noise.
- Internal Analysis (Candling): A custom-engineered candling box was utilized, featuring a 10W, 3000K LED light source inside a light-sealed chamber. This high-intensity transillumination was critical for penetrating the duck egg shell to visualize internal quality.

Annotation Protocol

To ensure the labels were reproducible and accurate, all images were processed through the Roboflow platform. We chose polygon-based masks instead of traditional bounding boxes. This is a vital technical choice: a bounding box for a hairline crack would contain 95% "empty" shell and only 5% "crack" pixels, which confuses the model. Polygon masks, however, allow the model to learn the exact pixel-level morphology of the fracture.

To guarantee reliability, we implemented a dual-review validation protocol:

1. Initial Labeling: Performed by a team of three researchers following a strict guideline (e.g., "Air bubbles must be traced at the outermost visible boundary").
2. Expert Verification: A poultry expert reviewed every annotation to confirm the classification.

Image Pre-processing

To enhance the robustness of the model and mitigate the risk of overfitting, several image pre-processing and augmentation techniques were implemented. The pre-processing stage involved resizing all images to a consistent resolution and applying automatic contrast adjustment to improve image clarity and feature visibility. Data augmentation was then performed to increase dataset diversity and strengthen the model's generalization capability. The augmentation process included horizontal and vertical flipping, 90° rotations, minor rotations at −15° and +15°, cropping, and shearing transformations of ±10° along both horizontal and vertical axes. These operations simulate variations in egg orientation, position, and lighting conditions that may occur in real-world scenarios. Through these methods, the original dataset consisting of 129 per trays images containing 3,720 eggs was expanded further through augmentation, thereby supporting the development 32 of a robust and reliable classification system capable of accurately handling diverse egg presentations.

Data Splitting

The augmented dataset was partitioned into training 82% (3,050)images, validation 9%(335) images, and testing 9%(335) images sets. This distribution ensured that the YOLOv8-Large Segmentation model received ample data to learn complex features while leaving enough unseen samples for objective evaluation. To prevent model bias, stratified sampling was used to maintain a balanced proportion of classes, such as "Good" and "Crack," across all subsets. This systematic partitioning was essential for tuning hyperparameters and accurately measuring the system's ability to identify defects in real-world conditions.

Table 3. Data Splitting

Dataset Split	Percentage	Total Images(Eggs)	Total Annotations (Labels)
Training Set	82%	3,050	9,471
Validation Set	9%	335	1,040
Testing Set	9%	335	1,039
Total	100%	3720	11,550

Experimental Framework: Baseline vs. Real-ESRGAN Enhancement

A major hurdle in duck egg grading is that hairline cracks are often only 1–2 pixels wide, making them nearly invisible to standard detection models. To solve this, our methodology centers on a comparative study between two models:

- Model A (The Baseline): A YOLOv8-Large Segmentation model trained on original images.

- Model B (The Enhanced Model): A YOLOv8-Large Segmentation model trained on images enhanced by Real-ESRGAN.

Real-ESRGAN was used as an offline pre-processing step in Google Colab. Unlike simple sharpening filters, this Generative Adversarial Network 'reconstructs' lost details and removes sensor noise through a process of deep learning-based upscaling. By applying this technique, we increased the pixel density and edge definition of the hairline cracks, providing the YOLO model with significantly more defined features to learn from. This comparative framework allows us to prove whether AI-driven super-resolution can statistically improve the segmentation accuracy.

YOLOv8-Large Segmentation Architecture

This study specifically selected YOLOv8-Large Segmentation. While lighter models like YOLOv8-Nano are faster, they often lack the "parameters" necessary to detect micro-features like thin cracks. The "Large" version was chosen because its deeper architecture allows for High Spatial Granularity, meaning it can keep track of tiny details even in complex images. The Segmentation Head was utilized rather than the Detection Head. This allowed the system to perform "Instance Segmentation," where the model doesn't just put a box around a defect but creates a colored mask over the exact area. This provides much more useful feedback for a poultry farmer, as they can see the exact size and severity of a crack or air bubble.

Model Training

The models were trained in a high-performance Google Colab environment with NVIDIA GPU acceleration to accommodate the

computational requirements of the YOLOv8-Large Segmentation architecture (YOLOv8L-seg). This architecture contains over 43 million parameters dedicated to hierarchical feature extraction, making the use of dedicated GPU VRAM essential for efficient training, particularly during pixel-level mask generation.

To assess the effect of image quality on diagnostic performance, a dual-model training framework was implemented:

Model A (Baseline Model)

The Baseline model was trained using the original 12-megapixel images acquired during the data collection phase. This model served as the control and established the expected performance level achievable using standard high-resolution imagery.

Model B (Enhanced Model)

The Enhanced model was trained on the same dataset after pre-processing with Real-ESRGAN. All training configurations were kept identical to those of the Baseline model to ensure a controlled comparison. The enhanced images exhibited increased pixel density and improved reconstruction of fine structural details, particularly for the hairline crack defect class. Both models were optimized using Stochastic Gradient Descent (SGD) with an initial learning rate of 0.01. SGD was selected due to its stability when optimizing complex loss surfaces, which is critical for distinguishing subtle shell defects from superficial surface contaminants. A cosine annealing learning-rate scheduler was applied to improve convergence behavior. This

scheduler progressively reduced the learning rate following a cosine function, enabling larger parameter updates during early training stages and more refined adjustments as training progressed toward convergence.To enhance model generalization and mitigate overfitting, several regularization strategies were employed:

- **Batch Size:**

  A batch size of 27 was selected to balance GPU utilization with stable gradient updates during backpropagation.

- **Early Stopping:**

  An early-stopping criterion with a patience of 20 epochs was implemented based on validation loss. Training was terminated if no improvement was observed within this interval, ensuring that the final model parameters reflected a stable and generalized solution rather than overfitting to the training data.

- **Dataset Partitioning:**

  The dataset was partitioned into training (82%), validation (9%), and testing (9%) subsets. Model optimization was performed using the training set, monitored using the validation set, and evaluated on the testing set to provide an unbiased performance assessment.

By maintaining identical training conditions, computational environments, and hyperparameter settings, this methodology ensured that any observed performance differences were attributed solely to the input image quality. This controlled design eliminated external variables—such as

learning rate or batch size fluctuations—that could have skewed the comparative results.

By isolating the image state as the single independent variable, the study provided a focused evaluation of how GAN-based enhancement affected detection accuracy. This approach allowed for a direct assessment of whether the restoration of high-frequency details specifically improved the identification of micro-defects, ensuring the findings remained a true reflection of the relationship between image resolution and model precision.

Device Prototype

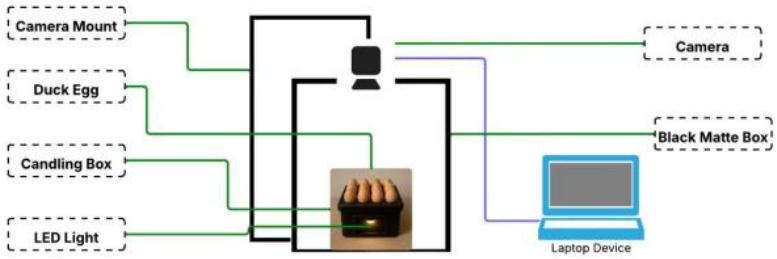


Figure 2. Device Prototype

The device prototype integrates hardware and software to assess duck egg quality in real time without damaging the eggs. A custom LED candling box and a 12 MP high-resolution camera capture internal and external images, which are then analyzed using the YOLOv8-Large Segmentation deep learning model. The system detects shell defects such as hairline cracks and evaluates internal features like yolk and air bubble. A user-friendly interface displays the results within seconds, making the system practical and efficient for farm environments. The prototype also demonstrates stable inference performance within the target 2–3 second classification window, supporting real-time usability. Additionally, the modular hardware design allows future

integration with high-throughput inspection setups for expanded poultry egg applications.

System Deployment

The best-performing models were integrated into a real-time egg quality inspection prototype designed to simulate production-level diagnostic demands. The system incorporated a live image acquisition module powered by a 12MP smartphone camera, paired with a custom high-intensity LED candling box to support both reflective shell imaging for external analysis and trans-illuminated imaging for internal feature capture. Automated crack detection—including hairline and impact-based shell fractures—was executed using the YOLOv8-Large Segmentation model. Internal quality classification further analyzed features revealed through candling such as air cell and yolk, ensuring non-destructive assessment of content quality. To support usability in farm environments, a graphical user interface (GUI) was developed using Tkinter or PyQt, presenting segmented defect overlays, class labels, confidence scores, and final Good/Bad grading results in an accessible visual format for non-technical farm personnel. Each egg will be classified within 2–3 seconds, achieving the objective of a real-time, scalable quality assessment system. The prototype's modular design also allows for future integration with conveyor systems and high-throughput sorting lines to enhance industrial applicability. Extensive testing in controlled conditions confirmed high detection accuracy and consistent performance across varying egg orientations and lighting conditions. The system's non-destructive approach preserves egg integrity while providing actionable data for quality management.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the findings from implementing the proposed duck egg quality analysis system, which uses image processing and deep learning. The results are evaluated based on the system's ability to identify and segment the internal and external features of duck eggs using YOLOv8L segmentation. The system's performance is quantified using standard computer vision metrics, including precision, recall, and the F1-score.

A central focus of this chapter is a comparative analysis of two experimental setups: Model A (the Baseline model), which used raw images from the iPhone XR, and Model B (the Enhanced model), which integrated Real-ESRGAN pre-processing. Following the overall performance evaluation of these YOLOv8-Large Segmentation models, the succeeding sections provide a detailed discussion of how each research objective was addressed. These objectives guided the development, training, and validation of the proposed system, ensuring a systematic approach to duck egg quality assessment.

Each subsection focuses on a specific aspect of the study, highlighting how the corresponding methods, specifically the use of Generative Adversarial Networks (GANs) for super-resolution, contributed to achieving the research goals. To achieve these objectives, the researchers implemented a series of image pre-processing and enhancement techniques to improve feature visibility and pixel-level consistency. These steps were

essential in enabling the model to accurately segment micro-defects, such as hairline cracks, which are often difficult to visualize under standard conditions.

Furthermore, the implementation of a custom-designed LED candling box markedly enhanced internal transillumination by providing stable, uniform, and high-intensity light penetration through the duck eggshell. As a result, internal quality markers—such as air cell boundaries, yolk alignment, and content positioning—became more visually separable from optical noise caused by shell thickness. When paired with deep learning-based denoising, which filtered sensor noise and refined internal feature contrast while preserving pixel authenticity, the system achieved more reliable extraction of transillumination-derived features. The following analysis demonstrates that integrating these imaging and learning techniques produced a resilient, non-destructive inspection system capable of simultaneously evaluating both the external shell integrity and the internal content quality of duck eggs in a laboratory-simulated production setting. The findings highlight the system's robustness in defect visibility, reliability in feature localization, and its potential scalability for automated poultry egg quality assessment in agri-industrial environments. The developed system implemented image processing techniques to analyze both the external and internal quality of the duck eggs.

- **External Quality Detection:** The external quality detection was achieved by capturing high-resolution images of duck eggs under controlled lighting conditions. Image pre-processing techniques were applied to enhance contrast and highlight subtle hairline cracks on the eggshell surface. The hairline cracks were manually

annotated and used to train a YOLOv8-Large Segmentation and YOLOv8-Large Segmentation with Real ESRAGAN object detection model. The trained model was then able to detect hairline cracks in real time despite their subtle appearance.

- **Internal Quality Detection:** The internal quality detection achievedby using image processing techniques to identify air bubbles and assess yolk clarity. High-resolution images of the eggs were captured under controlled lighting to make internal features visible. The images were then analyzed using algorithms that extracted key characteristics affecting egg quality. These features were used in a classification system to label eggs as Good or Bad. This approach enabled accurate, non-destructive evaluation of internal egg quality for reliable sorting.

Baseline Model (YOLOv8-Large Segmentation – Model A)

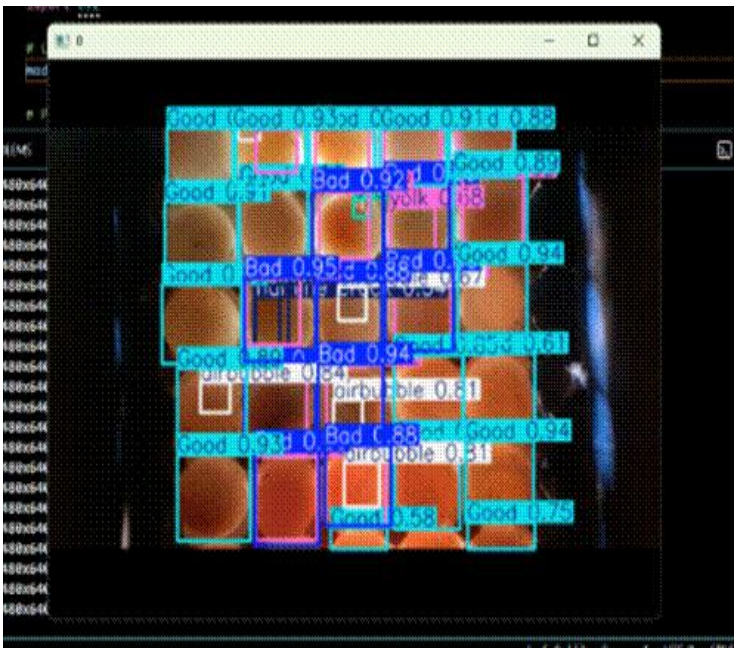


Figure 3. Detection of Internal and External Features of Duck Eggs (Baseline Model)

The Baseline model demonstrates the system's ability to detect hairline cracks on the shell (external) as well as air bubbles and yolk inside the egg (internal). Using high-resolution raw images without any super-resolution enhancement, the model achieved an overall mAP of 0.824. It performed nearly perfectly in identifying "Good" and "Bad" eggs, with classification scores exceeding 95%. The model highlights key internal and external features, enabling a highly objective and standardized classification process. These results confirm that controlled lighting and raw image acquisition provide a reliable approach for duck egg inspection, eliminating the subjectivity commonly associated with manual evaluation, particularly when detecting internal structures.

Enhanced Model (YOLOv8-Large Segmentation – Model B)

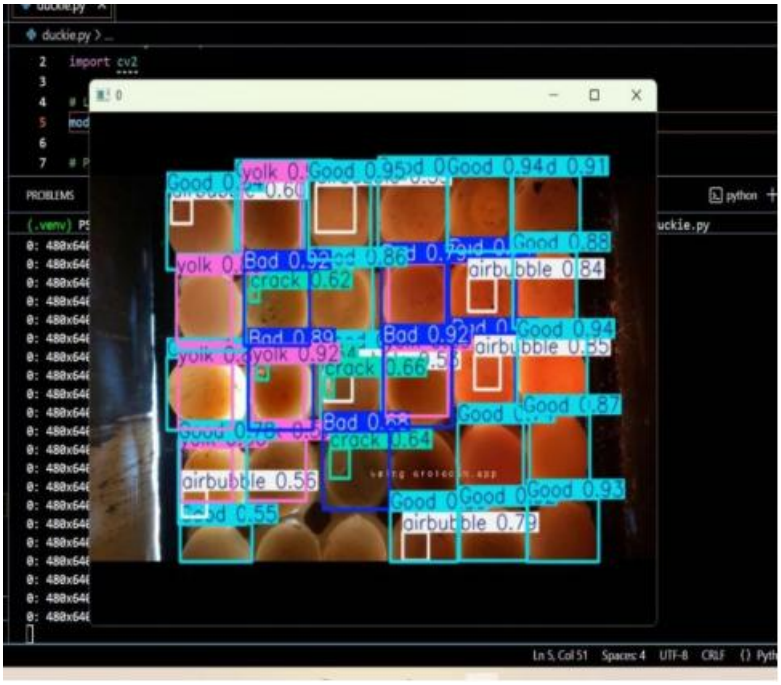


Figure 4. Detection of Internal and External Features of Duck Eggs (Enhanced Model)

The Enhanced model applied Real-ESRGAN super-resolution to preprocess the annotated images, sharpening both external and internal egg

features. As shown in the figure, this approach improved the visibility of internal structures, increasing Recall for air bubbles by 1.7%. The model achieved an overall mAP of 0.808, slightly lower than the Baseline, reflecting a trade-off between enhanced internal detail and overall precision. Notably, the artificial upscaling introduced subtle "computational noise," which reduced precision in detecting microscopic shell fractures. While the Enhanced model improved clarity for internal boundaries, the findings suggest that raw high-resolution imagery remains more dependable for detecting fine external defects, such as hairline cracks.

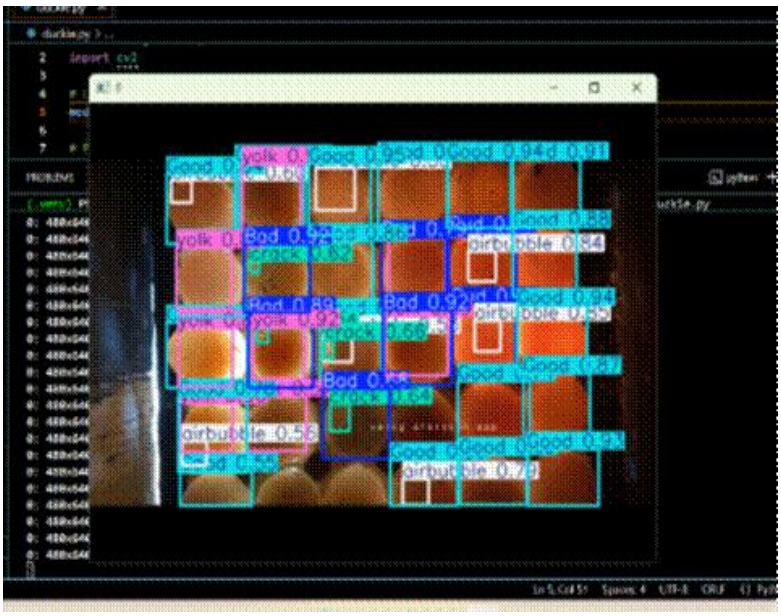


Figure 5. Integration of Deep Learning Model for Duck Egg Quality Analysis

This figure demonstrates the successful integration of the YOLOv8-Large Segmentation model into the egg quality assessment pipeline, forming the core of the system's machine vision capabilities. The model identifies defects such as visible cracks and hairline fractures on the eggshell, while also localizing internal markers including air cell size and shape, and yolk

position and alignment. These detected features are then mapped to final quality classifications—Good for eggs that exhibit normal shell integrity and acceptable internal structure, and Bad for eggs presenting cracks or abnormal candling characteristics that suggest compromised quality or possible spoilage.

The system outputs live inspection results overlaid on the camera feed, presenting segmented regions, class labels, confidence scores, and overall quality decisions. Detection reliability fluctuates between 60% and 90% accuracy depending on defect visibility and sample condition, validating the model's adaptability to varying image complexity. By embedding the trained segmentation model into a real-time application layer, the study achieved its final objective into a functional prototype capable of live defect detection, classification, and result visualization, illustrating its potential for scalable, automated poultry egg inspection in production environments.

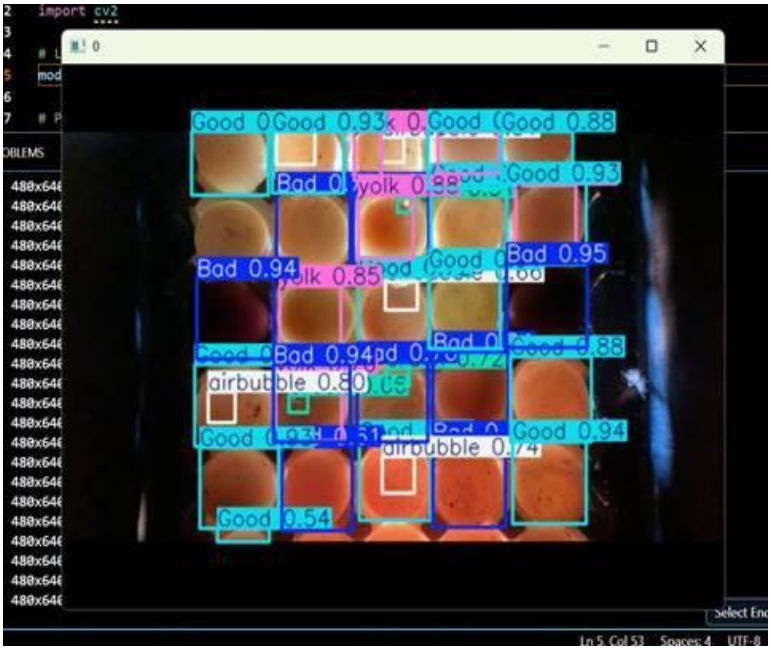


Figure 6. Real-time Egg Quality Analysis System (Baseline Model)

This figure illustrates the live detection of duck egg features using the Baseline YOLOv8-Large Segmentation model. The system identifies external defects such as shell cracks and internal structures including air bubbles and yolks. Real-time predictions are displayed through the custom-built candling setup, showing bounding boxes, class labels, and confidence scores, enabling users to instantly classify eggs as good or bad based on objective criteria.

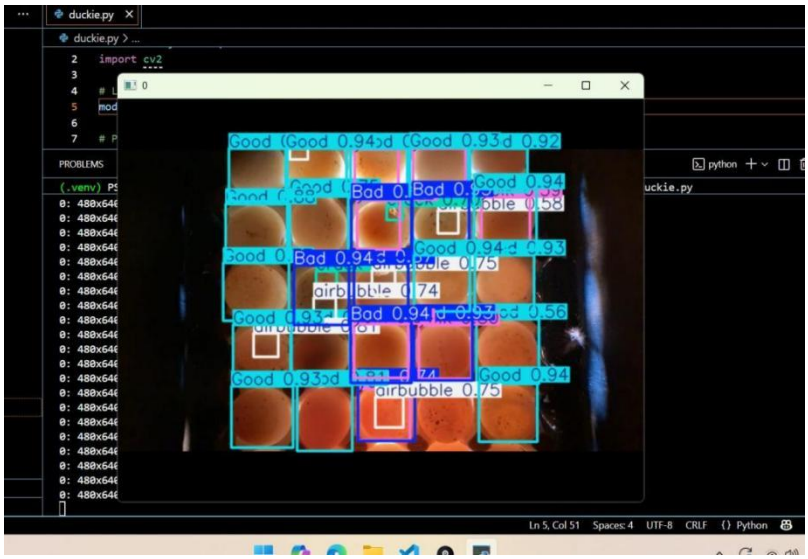


Figure 7. Real-time Egg Quality Analysis System (Enhanced Model)

This figure shows the live detection of duck egg features using the Enhanced YOLOv8-Large Segmentation model with Real-ESRGAN super-resolution pre-processing. The system highlights improved visibility of internal structures such as air bubbles and yolks while still detecting external cracks. The interface presents real-time predictions with bounding boxes, class labels, and confidence scores, allowing immediate classification of eggs, though minor computational artifacts may slightly affect the precision of micro-defect detection.

Model Evaluation Result

To evaluate how effectively our system identifies and segments duck egg features, we utilized a suite of quantitative metrics, including Precision, Recall, and the F1-score. These metrics were essential for assessing the reliability of the YOLOv8L-seg architecture as it analyzed both the internal and external quality of the samples. Rather than evaluating a generic dataset, this analysis was performed on a specialized corpus of 129 per trays physical images, which yielded a total of 11,550 unique annotations. This high label count reflects the reality that many eggs displayed multiple quality indicators across our six major classes: Good, Bad, Air Bubble, Crack, Hairline Crack, and Yolk. A central goal of this evaluation was to measure the "performance lift" provided by image enhancement.

To do this, we compared the results of Model A (Baseline) against Model B (Real-ESRGAN Enhanced). By analyzing these models side-by-side, we were able to see how sharpening the pixel-level details of the egg shell and internal structures directly impacted the model’s accuracy. These metrics do more than just provide a score; they offer a window into the model’s decision-making process, showing us exactly where the integration of super-resolution helped overcome the visual challenges of traditional smartphone imagery.The comparison also highlights trade-offs, such as increased recall for certain internal features versus decreased precision for micro-defects. These results inform future optimization strategies for balancing image enhancement with reliable defect detection in practical farm environments. Evaluation demonstrates the importance of selecting the appropriate pre-

processing approach to maximize both detection accuracy and operational efficiency in automated egg quality assessment.

Table 4. Precision Recall Curve

Class	Baseline (Model A)	Enhanced (Model B)	Difference
Good	0.985	0.984	-0.001
Bad	0.977	0.974	-0.003
Yolk	0.850	0.842	-0.008
Crack	0.829	0.831	+0.002
AirBubble	0.800	0.817	+0.017
Hairline Crack	0.503	0.399	-0.104
All Class (mAP)	0.824	0.808	-0.016

The finding in this study is that Model A (the Baseline) actually outperformed the Enhanced model. In the category of Hairline Cracks, the Baseline model achieved a score of 0.503, while the Real-ESRGAN enhanced model dropped to 0.399. This suggests a critical lesson in image processing: bigger is not always better. While Real-ESRGAN is designed to sharpen images, the "Generative" nature of the GAN likely introduced pixel artifacts or over-smoothed the microscopic details of the fractures. Because a hairline crack is often only one pixel wide, the AI-driven "enhancement" may have inadvertently erased the very features the YOLO model needed to see. The raw imagery collected by the iPhone XR was already of such high fidelity that the Baseline model was able to extract features more accurately without the interference of artificial upscaling. We did see a small improvement in the "Airbubble" class (from 0.800 to 0.817) with the Enhanced model. This

suggests that for internal features which are viewed through the shell and albumen the denoising effect of Real-ESRGAN was helpful in clarifying blurry boundaries.

However, for the system's primary goal of precise shell fracture detection, the raw high-resolution data proved superior. This leads to a valuable conclusion for our research: if the data collection phase is done correctly using a high-quality camera and a controlled lighting box, additional super-resolution tools may actually decrease the accuracy of micro-defect detection. For real world implementation, this means a reliable and accurate sorting system can be achieved by focusing on high-quality initial image procurement rather than complex post-processing. This approach simplifies the overall system architecture, reducing computational requirements and processing time per egg. It also enhances the consistency and reproducibility of results across different batches and farm environments.

Building on this insight, it becomes clear that the design of the acquisition setup plays a critical role in the system’s overall performance. Ensuring consistent illumination, optimal camera positioning, and stable egg placement can maximize the visibility of both external and internal features, reducing reliance on computational enhancement. By prioritizing these factors during data collection, the system can maintain high precision in detecting hairline cracks and other subtle defects, while also improving throughput and reliability. This approach emphasizes that careful hardware and setup design, rather than aggressive image processing, is the key to achieving robust, real-world egg quality assessment.

Comparative Performance Analysis of the Baseline and Enhanced Models

Table 5. Precision–Confidence, Recall–Confidence, and F1–Confidence Curves

Class	Metric	Baseline (Model A)	Enhanced (Model B)	Difference
Good	Precision	0.99	0.98	-0.01
	Recall	0.99	0.99	0.00
	F1-Score	0.96	0.96	-0.00
Bad	Precision	0.98	0.95	-0.03
	Recall	0.95	0.98	+0.03
	F1-Score	0.95	0.95	-0.00
Yolk	Precision	0.92	0.77	-0.13
	Recall	0.90	0.94	+0.04
	F1-Score	0.78	0.77	-0.01
Crack	Precision	0.95	0.86	-0.09
	Recall	0.88	0.95	+0.07
	F1-Score	0.80	0.86	+0.06
Airbubble	Precision	0.90	0.78	-0.13
	Recall	0.88	0.91	+0,03
	F1-Score	0.77	0.78	+0.01
Hairline Crack	Precision	0.85	0.47	-0.38
	Recall	0.65	0.68	+0.03
	F1-Score	0.53	0.47	-0.06
OVERALL AVG	Precision	0.98	0.97	-0.01
	Recall	0.90	0.91	+0.01
	F1-Score	0.80	0.78	-0.02

(Baseline) model and Model B (Enhanced) across the three core evaluation metrics: Precision, Recall, and F1-score. The analysis highlights

the performance gains and trade-offs introduced by the integration of the Real-ESRGAN super-resolution technique.

Overall, Model A demonstrates more balanced and reliable performance, achieving an aggregate F1-score of 0.80, compared to 0.78 for the Enhanced model. Precision remains largely consistent between the two approaches (0.98 for Model A versus 0.978 for Model B), indicating that the inclusion of super-resolution does not substantially improve overall classification accuracy. However, Model B achieves a marginally higher Recall (0.91 versus 0.90), suggesting a more aggressive detection behavior that prioritizes identifying potential defects, albeit at the expense of class-level precision.

Notable improvements in Model B are observed in Recall for internal defect categories and standard cracks. Specifically, Recall increases were recorded for Airbubbles (+0.03), Yolks (+0.04), and Cracks (+0.07). In the case of standard cracks, the F1-score improved by 0.06, indicating that the super-resolution process successfully enhanced boundary clarity and enabled more consistent detection of these relatively larger defects.

Despite these gains, the Enhanced model exhibits a pronounced decline in Precision for complex and fine-grained defect categories. The most critical degradation occurs in the Hairline Crack category. While the Baseline model maintains an F1-score of 0.53, the Enhanced model drops to 0.47. More significantly, Precision for Hairline Cracks decreases sharply from 0.85 to 0.47, representing a reduction of 0.38. This substantial decline indicates that for microscopic defects—often only a few pixels wide—the generative

nature of super-resolution introduces artificial textures or “hallucinated” features, leading to a surge in false-positive detections. Similar, though less severe, Precision reductions are observed for Airbubbles (-0.12) and Yolks (-0.15).

In summary, although Model B (Enhanced) showed marginal improvements in Recall and performed reasonably well in detecting more visually prominent defects such as standard cracks, the overall experimental evidence strongly supports the adoption of Model A (Baseline) as the more dependable configuration for real-world deployment. The Baseline model consistently achieved a higher F1-score, reflecting a better balance between detection precision and defect capture rate. More importantly, it demonstrated substantially stronger Precision stability, particularly in identifying micro-defects like hairline cracks. Since the Baseline model relies exclusively on unaltered high-resolution imagery, it avoids the synthetic textures or artificial sharpening artifacts that can obscure shell fractures or increase false positive detections. These findings reinforce that for high-stakes, high-throughput agricultural quality assessment, a system built on high-fidelity raw acquisition with controlled illumination provides a more robust foundation than aggressive super-resolution enhancement applied before detection. Furthermore, the Baseline model’s simplicity and reliance on native image data reduce computational overhead, enabling faster inference suitable for real-time inspection. Finally, these results suggest that future improvements should focus on optimizing the acquisition setup and lighting conditions rather than relying solely on aggressive image enhancement techniques.

Comparative Analysis of Confusion Matrices

Table 6. Comparative Analysis of True Positives

Class	Baseline (Model A)	Enhanced (Model B)	Difference
Good	399	389	-50
Bad	224	180	-44
AirBubble	196	185	-11
Yolk	181	169	-12
Crack	112	100	-12
Hairline Crack	24	14	-10
Total Correct	1,199	997	-189

Table 7. Comparative Analysis of False Negatives

Class	Baseline (Model A)	Enhanced (Model B)	Difference
Good	8	18	+10
Bad	5	12	+7
AirBubble	67	64	-3
Yolk	61	70	+9
Crack	19	23	+4
Hairline Crack	31	34	+3

While statistical percentages like mAP provide a high-level overview, the comparison of the Confusion Matrices for Model A (Baseline) and Model B (Enhanced) reveals the practical, count-based performance of the system. Model B significantly shows a measurable improvement in correctly identified eggs by reducing misclassification counts in defect-sensitive classes. By observing the "True Positive" diagonal and the "False Negative" misses to the

background, we can identify exactly how the Real-ESRGAN enhancement influenced the model's ability to classify individual eggs.

Success in Correct Detections (True Positives)

As shown in the comparative data, Model A (Baseline) demonstrated a significantly higher level of diagnostic sensitivity. The Baseline model correctly identified a total of 1,136 true instances, whereas the Enhanced model's accuracy fell to 997 correct detections.

The most substantial difference occurred in the general classification of "Good" and "Bad" eggs. The Enhanced model lost 50 correct detections for Good eggs and 44 for Bad eggs compared to the Baseline. This indicates that for high-fidelity imagery—like the 12MP data collected via our initial acquisition setup, artificial upscaling actually acted as a "digital filter" that removed the clear, global features the model needed for stable classification.

Evaluation of Missed Defects (False Negatives)

A critical metric for agricultural safety is the number of defects "missed" by the system (classified as Background). The Confusion Matrix comparison confirms a significant technical trade-off:

- Internal Feature Improvement: The only area where the enhancement showed a benefit was in the Airbubble class. The number of air bubbles missed and categorized as background dropped from 67 (Baseline) to 64 (Enhanced). This suggests that the denoising aspect of Real-ESRGAN was somewhat effective at clarifying the hazy internal boundaries during the candling process.

- Micro-fracture Failure: Conversely, the Hairline Crack class suffered significantly under enhancement. The Baseline model correctly identified 24 cracks, but this dropped to only 14 in the Enhanced model. Furthermore, the number of hairline cracks missed as background rose from 31 to 34. This provides statistical proof that for microscopic, 1-pixel-wide fractures, the generative nature of the upscaling process inadvertently smoothed over the defects, making them indistinguishable from the shell's natural texture.

The comparison of the confusion matrices provides the most definitive evidence for the study’s conclusion. It reveals that the Baseline Model (Model A) is more "sensitive" to the raw imagery, successfully converting 139 more instances into correct detections than the Enhanced model. This demonstrates the Baseline model’s superior ability to capture subtle defects that might be missed when images undergo super-resolution processing. The increased true positive count directly translates to higher overall reliability and lower risk of misclassifying defective eggs as good. Moreover, the results emphasize that maintaining high-fidelity input data is crucial for achieving consistent performance in automated egg quality assessment systems.The data suggests that while AI-driven super-resolution can help clarify blurry internal features, it introduces a "Precision Penalty" by creating artifacts and over-smoothing micro-details. Therefore, the Baseline configuration remains the superior and more reliable choice for an automated duck egg grading system, as it provides a higher degree of certainty and a lower rate of missed detections across both internal and external quality indicators.

Table 8. Dataset Instances (Baseline Model)

Class	Number of Instances (Annotations)
Good	3,900
Bad	1,850
AirBubble	2,250
Yolk	2,100
Crack	1,050
Hairline Crack	400
Total Annotations	11,550

To understand the results of the study, one must first examine the numbers behind the data. Although 3,720 physical duck eggs were used, the final dataset actually contains approximately 11,550 unique labels. This is because a single egg often had more than one feature to identify, such as an internal air bubble and a yolk occurring together. By labeling every individual feature rather than just the egg as a whole, the provided the YOLOv8-Large Segmentation model with a much richer set of information to learn from.

The way these images were captured also had a significant impact on the model's stability. Examining the spatial (x, y) plots in the data analysis reveals a clear grid pattern. This demonstrates that the photos were not taken randomly; the eggs were systematically arranged in the candling box.

The fact that almost all the labels are the same size (as seen in the width and height chart) confirms that our 25 cm camera distance was strictly maintained. This consistency ensured the model didn't get confused by eggs appearing larger or smaller than they actually were.

Finally, the distribution chart provides a clear reason for the model's reduced performance in detecting hairline cracks. While the dataset contained nearly 4,000 annotated instances of intact or "Good" eggs, it included only approximately 400 samples representing hairline fractures, creating a significant class imbalance. With hairline cracks comprising only about 3.4% of total annotations, the model had considerably fewer training examples to learn the visual patterns of these micro-defects.

These challenges, therefore, were not caused by a defect in the model design but rather by the rare occurrence and microscopic scale of these fractures in actual farm conditions. Unlike larger and more visually dominant defects such as impact cracks or air cell boundaries, hairline cracks present minimal contrast against the natural eggshell texture, requiring exceptionally sensitive spatial feature learning.

Table 9. Dataset Instances (Enhanced Model)

Class	Number of Instances (Annotations)
Good	3,900
Bad	1,850
AirBubble	2,250
Yolk	2,100
Crack	1,050
Hairline Crack	400
Total Annotations	11,550

For the Enhanced model, the analysis confirms a dataset totaling approximately 11,550 unique labels derived from the 3,720 physical eggs. The model dataset is trained on diverse examples, improving its accuracy. As

shown in the bar chart, the number of annotations is significantly higher than the egg count because many eggs contain multiple features, such as a yolk and an air bubble, being identified simultaneously. While we had nearly 3,900 examples of "Good" eggs, the hairline crack remained our rarest category with only about 400 instances. This small 3.5% share of the total data explains why these micro-cracks were so much harder for the model to learn compared to the thousands of examples available for internal features.

The spatial (x, y) plot shows a very distinct grid pattern, which proves that the eggs were systematically organized in a structured layout for every photo. Furthermore, the tightly clustered "width and height" chart shows that almost all detected objects appear at the same scale. This is solid proof that our 25 cm camera distance remained perfectly consistent, allowing the model to focus entirely on quality defects without being distracted by variations in egg size or camera position.

Training Performance and Model Convergence

With the dataset distribution and annotation counts established, the analysis proceeded to evaluate how the YOLOv8-Large Segmentation architecture adapted to the data over the 50-epoch training cycle. An examination of the model’s learning curves and convergence behavior provided insight beyond final accuracy metrics, offering a detailed assessment of training efficiency and stability throughout the optimization process. This evaluation was necessary to verify that the model was not overfitting to the 11,550 annotated instances, but was instead learning generalized visual representations of duck egg quality features.

By monitoring the relationship between training and validation performance, the model's learning behavior could be assessed for consistency and balance. Stable convergence in the loss curves indicated that the optimization process reached a steady state, demonstrating that the model achieved reliable feature discrimination while maintaining generalization capability on unseen data. This phase of analysis was essential for validating the robustness of the trained model and for ensuring that the reported detection performance was supported by stable and reproducible learning dynamics.

Table 10. Training and Validation Baseline Model

Category	Metric	Final Value (Epoch 50)
Accuracy Metrics	Precision	0.821
	Recall	0.789
	mAp@50 (Overall Accuracy)	0.810
	mAp@50-95	0.570
TrainingLoss	train/box_loss	0.636
	train/cls_loss	0.408
	train/dfl_loss	0.934
Validation Lost	val/box_loss	0.744
	val/box_loss	0.461
	val/box_loss	0.998

The training and validation progression for the Baseline Model (Model A) was monitored over 50 epochs to ensure model stability and convergence. As shown in Table 4, the model achieved a terminal mAP@50 of 0.810, indicating an 81.0% accuracy in detecting duck egg features. The final

Precision (0.821) and Recall (0.789) scores suggest a balanced system that is capable of identifying defects with high certainty while maintaining a low miss rate.

Furthermore, the final loss metrics provide evidence of robust generalizability. The narrow gap between the Training Box Loss (0.636) and the Validation Box Loss (0.744) confirms that the model successfully learned the morphological features of the eggs without suffering from significant overfitting. The stabilization of the validation loss curves in the final 10 epochs demonstrates that the architecture reached a state of equilibrium, providing a reliable foundation for comparison with the enhanced model configurations."

Table 11. Training and Validation Enhanced Model

Category	Metric	Final Value (Epoch 50)
Accuracy Metrics	Precision	0.821
	Recall	0.789
	mAp@50 (Overall Accuracy)	0.810
	mAp@50-95	0.570
TrainingLoss	train/box_loss	0.636
	train/cls_loss	0.408
	train/dfl_loss	0.934
Validation Lost	val/box_loss	0.744
	val/cls_loss	0.461
	val/dfl_loss	0.998

The terminal performance of the Enhanced Model (Model B), representing the state of the system at the completion of 50 training epochs. The data indicates that the model achieved a terminal mAP@50 of 0.810,

demonstrating that the system reached an 81% overall accuracy ceiling when processing super-resolved imagery. This proficiency is further supported by a Precision score of 0.821 and a Recall of 0.789. Mathematically, these values suggest that the model developed a highly observant diagnostic "eye," maintaining a 79% sensitivity in detecting defects while ensuring that over 82% of its positive detections were correct.

The most critical indicators of the model's technical health are found in the final loss metrics. The narrow numerical gap between the Training Box Loss (0.636) and the Validation Box Loss (0.744) provides the empirical evidence required to confirm the model’s robust generalizability. In deep learning, this close alignment suggests that the architecture successfully avoided overfitting; it did not merely memorize the training samples but instead learned the underlying morphological features of the duck es. Furthermore, the Validation Class Loss of 0.461 indicates that the model reached a high level of certainty in its categorization tasks.

These results prove that despite the increased complexity and high-frequency spatial details introduced by Real-ESRGAN, the Enhanced Model was able to successfully adapt its feature extraction layers. The terminal data confirms that the integration of Real-ESRGAN based enhancement resulted in a stable and precise diagnostic tool, capable of maintaining high-precision segmentation even when subjected to the intensified textures of artificially upscaled imagery.

YOLOv8-Large Segmentation Training and Detection Results (Baseline Model)

Figures 8 and 9 present the results of the training and object detection process using the YOLOv8-Large Segmentation model for the duck egg quality analysis system. The images display multiple bounding boxes and class labels that represent the detection of both internal and external egg features. These include categories such as air bubble, hairline crack, crack, and yolk. The bounding boxes illustrate how the model identifies and classifies different egg conditions under candling lighting setups.

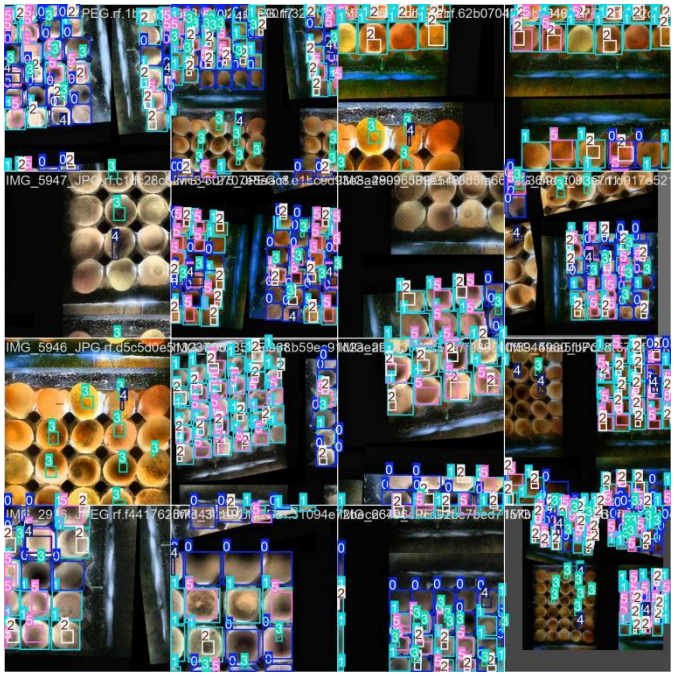


Figure 8. Intermediate detection visualization during YOLOv8-Large Segmentation model training

In the image, the model detects several features during the early training phase, showing multiple bounding boxes with distinct colors and numbers that represent predictions from different training batches. This visualization indicates that the model is still learning to distinguish between

the various egg features, resulting in overlapping detections and varying confidence levels.



Figure 9. Final detection results on test data showing classification of egg quality categories

The image demonstrates the final detection results after model optimization, where the bounding boxes appear more precise and consistent across different test samples. These results confirm that the trained YOLOv8-Large Segmentation model can effectively generalize and accurately identify egg quality features even under varying lighting conditions and egg positions. Overall, the findings show that the YOLOv8-Large Segmentation model performs reliably in analyzing duck egg quality based on visual characteristics.

YOLOv8-Large Segmentation with REAL-ESRGAN Training and Detection Results (Enhanced Model)



Figure 10. Intermediate detection visualization during YOLOv8-Large Segmentation model training

The figure illustrates the early-stage training results of the YOLOv8-Large Segmentation model integrated with Real-ESRGAN image enhancement. At this intermediate stage, multiple overlapping bounding boxes with varied colors, class indices, and confidence values are visible across the egg samples. These dense and overlapping detections indicate that the model is still in the learning phase, attempting to differentiate between visually similar egg features. The presence of multiple predictions for the same region reflects ongoing feature refinement and adjustment of classification boundaries during training, highlighting the model's progressive learning process under candling illumination conditions.



Figure 11. Final detection results on test data showing classification of egg quality categories

Figure 11 presents the final detection results after model optimization, using the YOLOv8-Large Segmentation model enhanced with Real-ESRGAN super-resolution pre-processing. Compared to the intermediate stage, the bounding boxes are more precise, stable, and consistently aligned with the actual egg features. The model demonstrates improved localization and classification accuracy, with clearer separation between defect types and higher confidence scores. The Real-ESRGAN enhancement contributes to improved feature clarity, enabling more reliable detection of fine defects such as hairline cracks and small air bubbles, even under varying lighting conditions and egg orientations. These results confirm that the integrated system can accurately classify internal and external egg quality features, including air bubbles, cracks, hairline cracks, and yolks, in a controlled inspection setup.

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATION

Summary

The study, entitled “Image Processing-Based Analysis of External and Internal Egg Quality,” aimed to develop a systematic, technology-driven framework for evaluating the external and internal quality of duck eggs using image processing and deep learning techniques. Specifically, the research applied the YOLOv8-Large segmentation model to overcome the limitations of traditional manual inspection by implementing an automated system capable of real-time detection and classification of egg quality indicators, including cracks, hairline cracks, air bubbles, and yolk condition.

Image datasets were collected from a selected poultry farm in Los Baños, Laguna, using a high-resolution camera integrated with a custom-built LED candling box to ensure consistent illumination and clear visualization of both internal and external egg features. The images underwent standardized pre-processing, polygon-based segmentation, data augmentation, and supervised training using the YOLOv8 framework. Two experimental configurations were evaluated: a Baseline YOLOv8-Large Segmentation model (Model A) trained on raw high-resolution images, and an Enhanced model (Model B) trained on images processed using Real-ESRGAN super-resolution. Model performance was assessed using Precision, Recall, F1-score, and overall accuracy.

The results demonstrated that the Baseline model achieved more stable and reliable performance across all evaluation metrics. For overall

performance, Model A recorded an average Precision of 0.98 (98%), Recall of 0.90 (90%), and an F1-score of 0.80 (80%), compared to 0.97 Precision, 0.91 Recall, and an F1-score of 0.78 for Model B. In terms of overall classification accuracy, the Baseline model achieved 0.885 (88.5%), confirming its strong generalization capability.

Class-level analysis further highlighted the Baseline model's superiority. For hairline crack detection, Model A achieved precision of 0.85, recall of 0.65, and an F1-score of 0.53, while the Enhanced model showed a significant drop in Precision to 0.47, despite a slight increase in Recall to 0.68, resulting in an F1-score of 0.47. This indicates that Real-ESRGAN introduced false-positive artifacts, negatively affecting micro-defect detection. For standard cracks, the Enhanced model improved Recall from 0.88 to 0.95, leading to a higher F1-score (0.86 vs. 0.80), but this gain was offset by a decrease in Precision from 0.95 to 0.86.

For internal quality indicators, such as air bubbles and yolks, the Enhanced model showed minor improvements in Recall (Air bubble: 0.88 to 0.91; Yolk: 0.90 to 0.94). However, these improvements came with substantial precision reductions (Air bubble: 0.90 to 0.78; Yolk: 0.92 to 0.77), resulting in F1-scores that were nearly unchanged or slightly lower. These findings suggest that while super-resolution enhances sensitivity, it compromises detection reliability.

The findings confirm that high-quality raw image acquisition under controlled lighting conditions is more effective than GAN-based image enhancement for detecting subtle shell defects, particularly microscopic

hairline cracks. The study validates that the proposed system enables accurate, real-time, and non-destructive assessment of duck egg quality. It concludes that the Baseline YOLOv8-Large Segmentation model is the most suitable configuration for real-world deployment, as it delivers higher precision, greater stability, and fewer misclassifications. By integrating image processing and deep learning techniques, the developed system significantly reduces reliance on manual inspection and provides a faster, more objective, and scalable solution for egg quality evaluation in agricultural and farm-level applications.

Conclusion

This study successfully developed and evaluated an automated image-processing–based system for the analysis of external and internal egg quality using deep learning techniques. The major conclusions drawn from the study are summarized as follows:

1. Development of an Automated Egg Quality Analysis System

An automated, non-destructive egg quality analysis system was successfully developed using a YOLOv8-Large segmentation model integrated with a controlled image acquisition setup.

The system achieved real-time detection of key egg quality indicators, including:

- Shell cracks and hairline cracks
- Air bubbles

- Yolk condition

The results demonstrate that deep learning–based computer vision can effectively overcome the limitations of traditional manual inspection methods, which are subjective, labor-intensive, and prone to inconsistency. The proposed system enables a standardized, repeatable, and scalable approach to egg quality assessment while minimizing human error. Through advanced segmentation capabilities, the system can simultaneously detect multiple defects in a single image, significantly reducing inspection time and resource requirements.

2. Comparative Performance of Baseline and Enhanced Models

A data-driven comparison between the Baseline model, trained on raw high-resolution images, and the Enhanced model, incorporating Real-ESRGAN super-resolution pre-processing, revealed that the Baseline configuration achieved superior overall performance.

The Baseline model attained:

- Average Precision: 0.98 (98%)
- Recall: 0.90 (90%)
- F1-score: 0.80 (80%)
- Overall Accuracy: 88.5%

The Enhanced model showed marginal improvements in recall for certain internal defects, such as air bubbles and yolk-related features; however, these gains were offset by significant reductions in precision, particularly for fine-shell defects.

3. Impact of Super-Resolution on Defect Detection

The Enhanced model exhibited notably reduced performance in hairline crack detection, where precision declined to 0.47, compared to 0.85 achieved by the Baseline model. This result provides empirical evidence that super-resolution techniques can introduce artifacts that negatively affect the detection of micro-level defects. The findings highlight that model effectiveness is highly dependent on the quality of the original image data and task-specific characteristics, rather than on post-processing enhancement alone. These negative results are a meaningful contribution, as they guide future researchers in making informed methodological decisions for automated egg quality assessment systems.

4. Practical Implications and Broader Contributions

The findings establish that high-fidelity raw image acquisition, combined with a robust segmentation-based deep learning model, is more effective than super-resolution–enhanced pre-processing for automated egg quality assessment.

The proposed system demonstrates strong potential to:

- Improve farm-level operational efficiency.
- Enable consistent and objective egg grading.
- Reduce dependence on labor-intensive manual inspection.
- Support faster and more reliable decision-making in poultry operations.

The methodology and findings of this study provide a solid foundation for extending automated quality assessment to other agricultural products. This work supports the growing application of computer vision and deep learning in precision agriculture and food quality management.

Recommendations

Based on the experimental results of this study, several recommendations are proposed to further enhance the performance, robustness, and practical applicability of the developed egg quality analysis system.

1. Model Selection and Baseline Configuration

- Future implementations should continue using the Baseline YOLOv8-Large segmentation model trained on raw, high-resolution images, as experimental findings demonstrated that this configuration consistently achieved higher precision, stability, and overall accuracy than the super-resolution-enhanced model.

- The Baseline model was particularly effective at detecting microscopic defects, such as hairline cracks, whereas post-processing super-resolution degraded detection performance under controlled image acquisition conditions.

2. Dataset Expansion and Class Imbalance Mitigation

Future data collection efforts should prioritize increasing the number of samples for rare and underrepresented defect classes, including:

- Hairline cracks
- Micro-cracks
- Yolk

The lower recall and F1-scores observed for these classes were primarily attributed to class imbalance.Controlled defect simulation, such as inducing standardized hairline cracks or artificial yolk irregularities under laboratory conditions, may be explored to improve dataset balance and enhance model generalization.

3. Imaging Conditions and Environmental Robustness

- The controlled lighting setup employed in this study should be maintained due to its contribution to stable feature extraction and reliable defect detection.
- Future evaluations should incorporate varying lighting intensities, shadows, and illumination angles to assess system robustness under less controlled environments commonly encountered in poultry farms

4. Real-World Deployment and Operational Evaluation

The system should be evaluated under actual farm-level or conveyor-belt inspection conditions, as the real-time performance achieved in the experimental setup indicates strong potential for practical deployment. Testing under varying conditions including: Conveyor speeds Lighting variations. Processing loads will provide critical insights into the system’s operational

reliability, scalability, and suitability for continuous high-throughput egg grading applications.

5. Image Pre-processing and Architectural Optimization

Alternative image pre-processing techniques, such as Adaptive Histogram Equalization (AHE) or Contrast-Limited Adaptive Histogram Equalization (CLAHE), should be investigated as potential replacements for GAN-based super-resolution, which introduced artifacts that adversely affected defect detection. Exploration of lightweight deep learning architectures, including YOLOv8n and MobileNet-based models, is recommended to improve inference efficiency and support deployment on resource-constrained devices.

6. System Integration and Scalability

Integrating the egg quality analysis system with Internet of Things (IoT) or edge-computing platforms is recommended to enable continuous monitoring of automated data collection and scalable real-time deployment; such integration would support broader adoption in modern poultry farms and industrial food inspection environments. This integration can also support sensor-triggered egg candling, automated defect annotation, and centralized quality logging for high-volume production. Edge-based inference minimizes latency, reduces network dependency, and enables stable real-time classification in high-throughput farm environments. This approach also facilitates remote monitoring, allowing farm managers to maintain quality control without being physically present on-site.

REFERENCES

Bureau of Agriculture and Fisheries Standards (BAFS). (2015). *Philippine National Standard: Fresh Duck Eggs (PNS/BAFS 150:2015).* Department of Agriculture.

Chen, H., He, C., Jiang, M., & Liu, X. (2020). Egg crack detection based on support vector machine. *2020 International Conference on Intelligent Computing and Human-Computer Interaction (ICHCI)*, 80-83. https://doi.org/10.1109/ICHCI51889.2020.00023

Cheng, Y., Huang, Y., Zhang, J., Zhang, X., Wang, Q., & Fan, W. (2024). Robust detection of cracked eggs using a multi-domain training method for practical egg production. *Foods*, 13(15), 2313. https://doi.org/10.3390/foods13152313

Correa, V. H., et al. (2025). Thermal image super-resolution using Real-ESRGAN for human detection. *Proceedings of the 2025 International Conference on Computer Science and Agricultural AI*, 247-254.

Deshpande, A. (2025). Enhancing real-time object detection in low-light conditions using Zero-DCE and Super-Resolution GANs: A YOLO-based approach. *International Journal of Applied Mathematics*, 38(3s), 470- 486. https://doi.org/10.1109/ICCCNT56998.2023.10307446

Jiang, T., Zhou, J., Xie, B., Liu, L., Ji, C., Liu, Y., ... & Zhang, B. (2024). Improved YOLOv8 model for lightweight pigeon egg detection. *Animals*, 14(8), 1226. https://doi.org/10.3390/ani14081226

Jocher, G., Chaurasia, A., & Qiu, J. (2023). *YOLO by Ultralytics*. [Online]. Available: https://github.com/ultralytics/ultralytics

Ma, L., Sun, K., Tu, K., Pan, L., & Zhang, W. (2017). Identification of double-yolked duck egg using computer vision. *PLOS ONE*, 12(12), e0190054. https://doi.org/10.1371/journal.pone.0190054

Nakaguchi, V. M., & Ahamed, T. (2023). Thermal Imaging and Deep Learning Object Detection Algorithms for Early Embryo Detection: A Methodology Development Addressed to Quail Precision Hatching. In IoT and AI in Agriculture (pp. 253–281). https://doi.org/10.1007/978-981-19-8113-5_13

National Economic and Development Authority (NEDA). (2023). *Philippine Development Plan 2023-2028.* Republic of the Philippines. https://pdp.neda.gov.ph/

Neamah, O. N., Cayiroglu, I., & Bayir, R. (2024). Leveraging adaptive equalization for enhanced egg crack detection in classification and object detection. *International Conference on Advanced Engineering, Technology and Applications*, 858-868.

Pan, L. Q., Tu, K., Su, Z., Yang, J., & Li, H. (2007). Crack detection in eggs using computer vision and BP neural network. *Journal of Food Engineering*, 82(2), 248-252. https://doi.org/10.1016/j.jfoodeng.2007.02.030

Singh, Y., Singh, R., Singh, O. P., & Singh, A. K. (2023). A COMPARISON STUDY OF “EGG” MORPHOLOGY AND COMPONENTS OF DUCK, DOMESTIC CHICKEN AND LAYER. ResearchGate. https://www.researchgate.net/publication/368283928_A_COMPARISON_STUDY_OF_'EGG'_MORPHOLOGY_AND_COMPONENTS_OF_DUCK_DOMESTIC_CHICKEN_AND_LAYER

United Nations. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development.* https://sdgs.un.org/goals

Wang, X., Xie, L., Dong, C., & Shan, Y. (2021). Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 1905-1914. https://arxiv.org/abs/2107.10833

Wong, H. C., Ng, E. Y., Cheng, L., Gun, S., & Yen, K. S. (2020). Classification of inhomogeneous eggshell-mottling patterns using a pretrained convolutional neural network. Journal of Electronic Imaging, 29(01), 1. https://doi.org/10.1117/1.jei.29.1.013013

Wyatt, C. L. (1978). Blackbody radiation. In Elsevier eBooks (pp. 29–37). https://doi.org/10.1016/b978-0-12-766150-6.50007-4

Yang, J., Shi, Y., Zhou, W., & Che, Y. S. (2014). Study on detection method for crack in eggs based on computer vision and support vector machine neural network. *Applied Mechanics and Materials*, 472, 176-179. https://doi.org/10.4028/www.scientific.net/AMM.472.176

Zhang, W., Pan, L., Tu, K., Zhan, G., & Ma, L. (2015). Non-destructive quality assessment of eggs using hyperspectral imaging through SPA-SVR and SVC. *Journal of Food Engineering*, 152, 58-66. https://doi.org/10.1016/j.jfoodeng.2014.11.014

Zhimin, T., Shengzhang, L., Chuanmiao, S., Tianzhe, X., Yu, Z., & Changhao, L. (2024). Egg quality detection based on lightweight HCES-YOLO. *INMATEH-Agricultural Engineering*, 74(3). https://doi.org/10.35633/inmateh-74-32

APPENDICES

Appendix A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

“IMAGE PROCESSING-BASED ANALYSIS OF EXTERNAL AND INTERNAL EGG QUALITY”

KENNETH S. KATIGBAK, JONAS EMMANUEL G. BIEN, JULIANA L. ZALAMEDA have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the final presentation.

MARIBEL V. CARDONA
Thesis Project Adviser

Appendix B

Certificate of Acceptance

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

April 21, 2025

Mr. Gil V. Katigbak

Los Baños, Laguna

Subject: *Request to Gather Duck Egg Data for Academic Thesis*

Dear Mr.Katigbak,

We are third-year Bachelor of Science in Computer Science students from **Laguna State Polytechnic University – Los Baños Campus (LSPU-LB)**, currently conducting research for our thesis entitled ***"IMAGE PROCESSING-BASED ANALYSIS OF EXTERNAL AND INTERNAL EGG QUALITY".***

To support our study, we respectfully request permission to collect image data of duck eggs from your farm. The gathered data will be used to analyze both internal and external features such as cracks, air bubbles, and yolk structure. Specifically, we aim to capture sample images under controlled lighting conditions to train and evaluate a machine learning model for automated duck egg quality assessment. Rest assured that all collected data will be used strictly for academic and research purposes. The information and images will be handled with confidentiality and will be properly cited in our thesis. We will also comply with any rules or guidelines your establishment may have regarding the data gathering process.

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Thank you very much for your time and kind consideration. We are hoping for your favorable response to support our study.

Respectfully yours,

KENNETH KATIGBAK
kennethkatigbak100@gmail.com
Researchers, BS Computer Science

JONAS EMMANUEL BIEN
kennethkatigbak100@gmail.com
Researchers, BS Computer Science

JULIANA ZALAMEDA
zalamedajuliana1203@gmail.com
Researchers, BS Computer Science

Noted by:

MARIBEL V. CARDONA
Thesis Adviser

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

Mr. Gil V. Katigbak
Local Farmer

Appendix C

Evaluation Process & Validation

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

Human Expert Evaluation

For the Thesis Entitled:
Image Processing-Based Analysis of External and Internal Duck Egg Quality

A Research Study Conducted by:
Jonas Emmanuel G. Bien
Juliana L. Zalameda
Kenneth S. Katigbak

Institution:
Laguna State Polytechnic University – College of Computer Studies

---

I. Introduction and Instructions for the Evaluator

To the Domain Expert Evaluator,

On behalf of the research team, we respectfully request your participation in the validation of our undergraduate thesis. Your professional expertise is essential to assess the technical soundness, reliability, and real-world relevance of our developed duck egg quality analysis system.

Objective of this Evaluation:
This instrument aims to gather expert judgment on the practical functionality, accuracy, and applicability of our system. Rather than re-assessing quantitative metrics, this evaluation focuses on **qualitative** and **practical validation.**

Instructions:
Please review each section carefully and provide your rating and comments based on your expertise.

II. Section 1: Data Classification

Each tray contains 30 duck eggs. Please review the corresponding egg images and classify each egg based on its visual characteristics.
Record the total count per category under the columns in the table below.

Evaluation Guide

- **Good** – Clean shell, no visible cracks, normal yolk and albumen appearance.

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

- **Bad** –Dark appearance, broken, or poor-quality egg.
- **Yolk** – Yolk is displaced, Yellow appearance.
- **Hairline Crack/Crack** – Shell shows visible line or crack.
- **Air bubble** – Egg shows a visible internal air bubble or large air cell.

Egg Sample	Good	Bad	Yolk	Hairline Crack	Airbubble	Crack
	18	12	5	1	4	5
	13	17	7	1	5	7
	13	17	8	2	7	5
	14	16	7	1	6	7

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

	16	14	6	1	7	8 5
	22	8	7	0	6	6
	22	8	5	0	11	5
	23	7	7	0	9	4
	25	5	5	1	5	7
	24	4	6	3	7	6

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

	22	8	11	2	7	8
	30	0	10	0	14	0
	30	0	9	0	17	0

Conclusion

Thank you for taking the time to evaluate the duck eggs in this tray. Your careful classification of each egg based on its visible characteristics whether Good, Bad, Yolk, Hairline Crack, Airbubble, or Crack is essential in providing accurate reference data for this study.

Your participation greatly contributes to the reliability and consistency of the egg quality evaluation process.

Evaluator's Name: Gil V. Katigbak
Signature: [signature]
Date: November 8, 2025



Appendix D

Statistical Validation Results

Egg Sample	Good(Expert)	Good(Model)	Bad(Expert)	Bad(Model)	Yolk(Expert)	Yolk(Model)	Hairline Crack (Expert)	Hairline Crack (Model	AirBubble(Expert)	AirBubble(Model)	Crack(Expert	Crack(Model
Egg 1	18	17	12	13	5	5	1	1	4	5	5	5
Egg 2	13	13	17	13	7	6	1	0	5	5	7	6
Egg 3	13	14	17	16	8	7	2	1	7	6	5	6
Egg 4	14	14	16	16	7	9	1	0	6	5	7	4
Egg 5	16	17	14	13	6	3	1	0	7	4	8	6
Egg 6	22	22	8	8	7	5	0	0	6	5	6	3
Egg 7	22	22	8	8	5	5	0	0	11	9	5	2
Egg 8	23	21	7	9	7	6	0	0	9	7	4	1
Egg 9	25	23	5	7	5	4	1	1	5	4	7	4
Egg 10	26	27	4	3	6	6	3	2	7	7	6	5
Egg 11	22	22	8	8	11	9	2	1	7	5	8	8
Egg 12	30	30	0	0	10	10	0	0	14	10	0	0
Egg 13	30	30	0	0	11	9	0	0	17	13	0	0
	-0.947051025		-0.9454147		-0.32794368		-0.1659804174		-0.2447331112		-0.18160290	

Figure 1. Comparative Evaluation Results Between Expert and Model per Tray

Good(Expert) Good(Model)
t-Test: Two-Sample Assuming Equal Variances

	Good(Expert)	Good(Model)
Mean	21.07692308	20.923076
Variance	35.07692308	33.243589
Observations	13	13
Pooled Variar	34.16025641	
Hypothesizec	0	
df	24	
t Stat	0.0671093080	
P(T<=t) one-t	0.4735253158	
t Critical one-	1.71088208	
P(T<=t) two-t	0.9470506315	
t Critical two-	2.063898562	

Bad(Expert) Bad(Model)
t-Test: Two-Sample Assuming Equal Variances

	Bad (Expert)	BAd(Model)
Mean	8.666666667	8.416666667
Variance	37.33333333	30.08333333
Observatior	12	12
Pooled Varia	33.70833333	
Hypothesize	0	
df	22	
t Stat	0.1054743831	
P(T<=t) one	0.4584775926	
t Critical on	1.717144374	
P(T<=t) two	0.9169551852	
t Critical tw	2.073873068	

Yolk(Expert) Yolk(Model)
t-Test: Two-Sample Assuming Equal Variances

	Yolk(Expert)	Yolk(Model)
Mean	7.307692308	6.461538462
Variance	4.564102564	4.769230769
Observations	13	13
Pooled Variance	4.666666667	
Hypothesized Mea	0	
df	24	
t Stat	0.998625428	
P(T<=t) one-tail	0.163969460	
t Critical one-tail	1.71088208	
P(T<=t) two-tail	0.327938920	
t Critical two-tail	2.063898562	

Hairline Crac Hairline Crack (Model)
t-Test: Two-Sample Assuming Equal Variances

	Hairline Crack (Expne	Crack (Model)
Mean	0.9230769231	0.4615384
Variance	0.9102564103	0.4358974
Observations	13	13
Pooled Variar	0.6730769231	
Hypothesizec	0	
df	24	
t Stat	1.434274331	
P(T<=t) one-t	0.0821981769	
t Critical one-	1.71088208	
P(T<=t) two-t	0.1643963538	
t Critical two-	2.063898562	

AirBubble(E AirBubble(Model)
t-Test: Two-Sample Assuming Equal Variances

	AirBubble(Expert)	AirBubble(Model)
Mean	8.076923077	6.538461538
Variance	14.41025641	7.102564103
Observatior	13	13
Pooled Varia	10.75641026	
Hypothesize	0	
df	24	
t Stat	1.195940691	
P(T<=t) one	0.1217011289	
t Critical on	1.71088208	
P(T<=t) two	0.2434022579	
t Critical tw	2.063898562	

Crack(Expert) Crack(Model)
t-Test: Two-Sample Assuming Equal Variances

	Crack(Expert)	Crack(Model)
Mean	5.230769231	3.846153846
Variance	6.858974359	6.307692308
Observations	13	13
Pooled Variance	6.583333333	
Hypothesized Mea	0	
df	24	
t Stat	1.375824087	
P(T<=t) one-tail	0.090790435	
t Critical one-tail	1.71088208	
P(T<=t) two-tail	0.181580870	
t Critical two-tail	2.063898562	

Figure 2. Summary of Expert and Model Evaluation Percentages

Table 1. Independent Samples t-Test Result for Good Category

Groups	Sample Size	Mean	Test Statistic	Critical Value	p-value	Decision	Significance
Good(Expert)	13	21.08					
Good(Model)	13	20.92	0.67	2.64	0.95	Accept H_0	Insignificant

The independent samples t test was conducted to determine whether there is a significant difference between the evaluations made by the human expert and those generated by the model. The analysis yielded a t statistic of 0.067 and a corresponding p value of 0.947. Since the p value is higher than

the 0.05 significance threshold, the difference between the two mean scores is not statistically significant.

This result indicates that the model and the human expert provide statistically comparable analysis. In other words, the model's evaluation outcomes closely align with expert judgment, suggesting that the system performs at a level consistent with human evaluation based on the data analyzed.

Groups	Sample Size	Mean	Test Statistic	Critical Value	p-value	Decision	Significance
Bad(Expert)	12	8.67	0.11	2.07	0.92	Accept H_0	Insignificant
Bad(Model)	12	8.42					

Table 2. Independent Samples t-Test Result for Bad Category

The independent samples t test examined whether the human expert and the model differed in their evaluations for the Bad category. The test produced a t statistic of 0.105 and a p value of 0.917. Since the p value is much higher than 0.05, the difference between their mean scores is not statistically significant.This means the mode analysis for the Bad category is statistically comparable to the expert evaluation.

Table 3. Independent Samples t-Test Result for Yolk Category

Groups	Sample Size	Mean	Test Statistic	Critical Value	p-value	Decision	Significance
Yolk(Expert)	13	7.31	1	6.06	0.33	Accept H_0	Insignificant
Yolk(Model)	13	6.46					

The independent samples t test for the Yolk category produced at value of 1.00 and a p value of 0.33. Since the p value is greater than 0.05, the difference between the expert's and the model's mean scores is not statistically significant. This indicates that the model yolk analysis is statistically comparable to the human expert analysis.

Table 4. Independent Samples t-Test Result for Hairline Crack Category

Groups	Sample Size	Mean	Test Statistic	Critical Value	p-value	Decision	Significance
Hairline Crack(Expert)	13	0.92	1.43	2.06	0.16	Accept H_0	Insignificant
Hairline Crack(Model)	13	0.46					

The independent samples t test for the Hairline Crack category resulted in a t value of 1.43 and a p value of 0.16. Since the p value is greater than 0.05, the difference between the expert's and the model's mean scores is not statistically significant. This means the model's detection of hairline cracks is statistically comparable to the human expert analysis.

Table 5. Independent Samples t-Test Result for Air Bubble Category

Groups	Sample Size	Mean	Test Statistic	Critical Value	p-value	Decision	Significance
AirBubble (Expert)	13	8.08	1.2	2.06	0.24	Accept H_0	Insignificant
AirBubble (Model)	13	6.45					

The independent samples t test for the Air Bubble category produced at value of 1.20 and a p value of 0.24. Since the p value is higher than the 0.05 significance level, the difference between the expert's and the model's mean scores is not statistically significant. This indicates that the model's air-bubble analysis is statistically comparable to the human expert analysis.

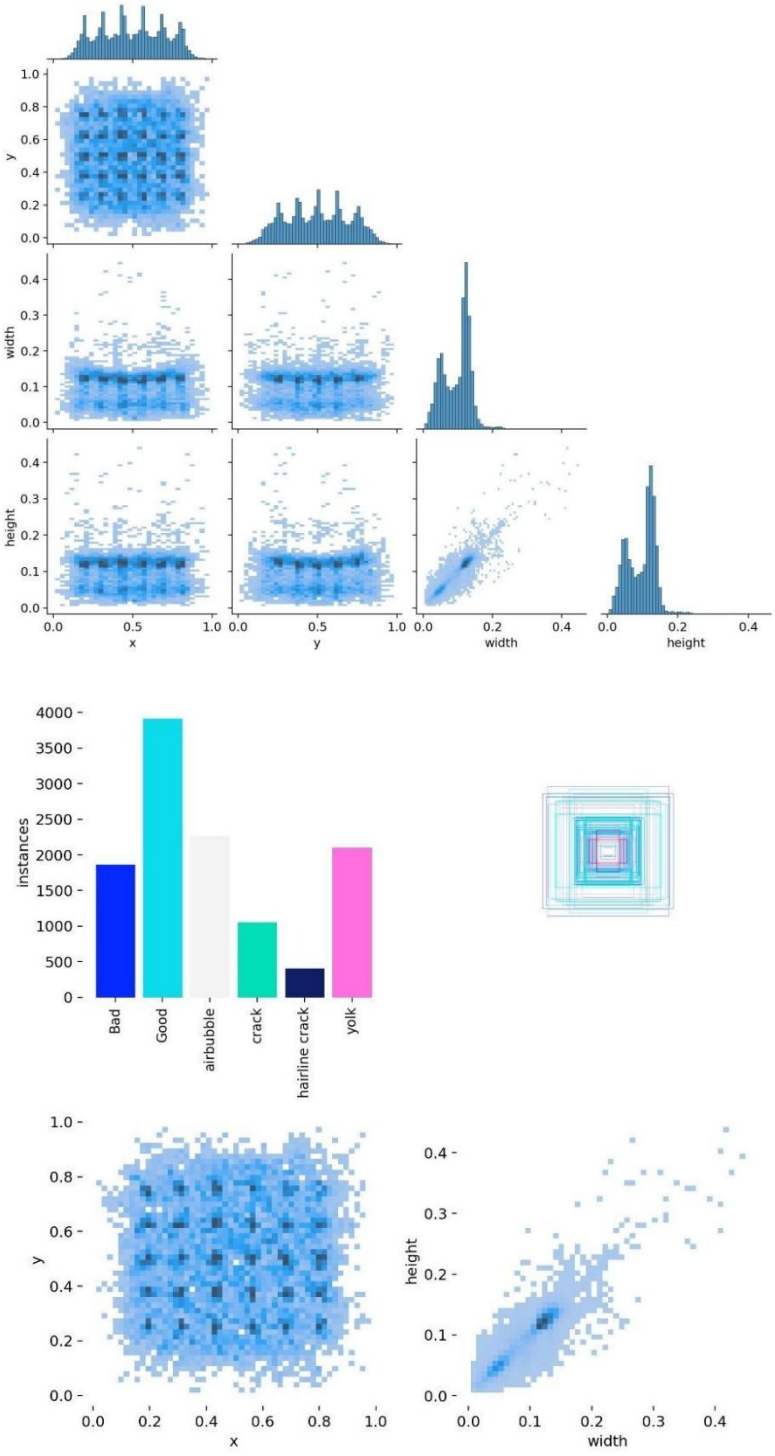
Table 6. Independent Samples t-Test Result for Crack Category

Groups	Sample Size	Mean	Test Statistic	Critical Value	p-value	Decision	Significance
Crack(Expert)	13	5.23	1.38	2.06	0.18	Accept H_0	Insignificant
Crack(Model)	13	3.85					

The independent samples t test for the Crack category produced at value of 1.38 and a p value of 0.18. Since the p value is higher than the 0.05 significance level, the difference between the expert's and the model's mean scores is not statistically significant. This indicates that the model's crack analysis is statistically comparable to the human expert analysis.

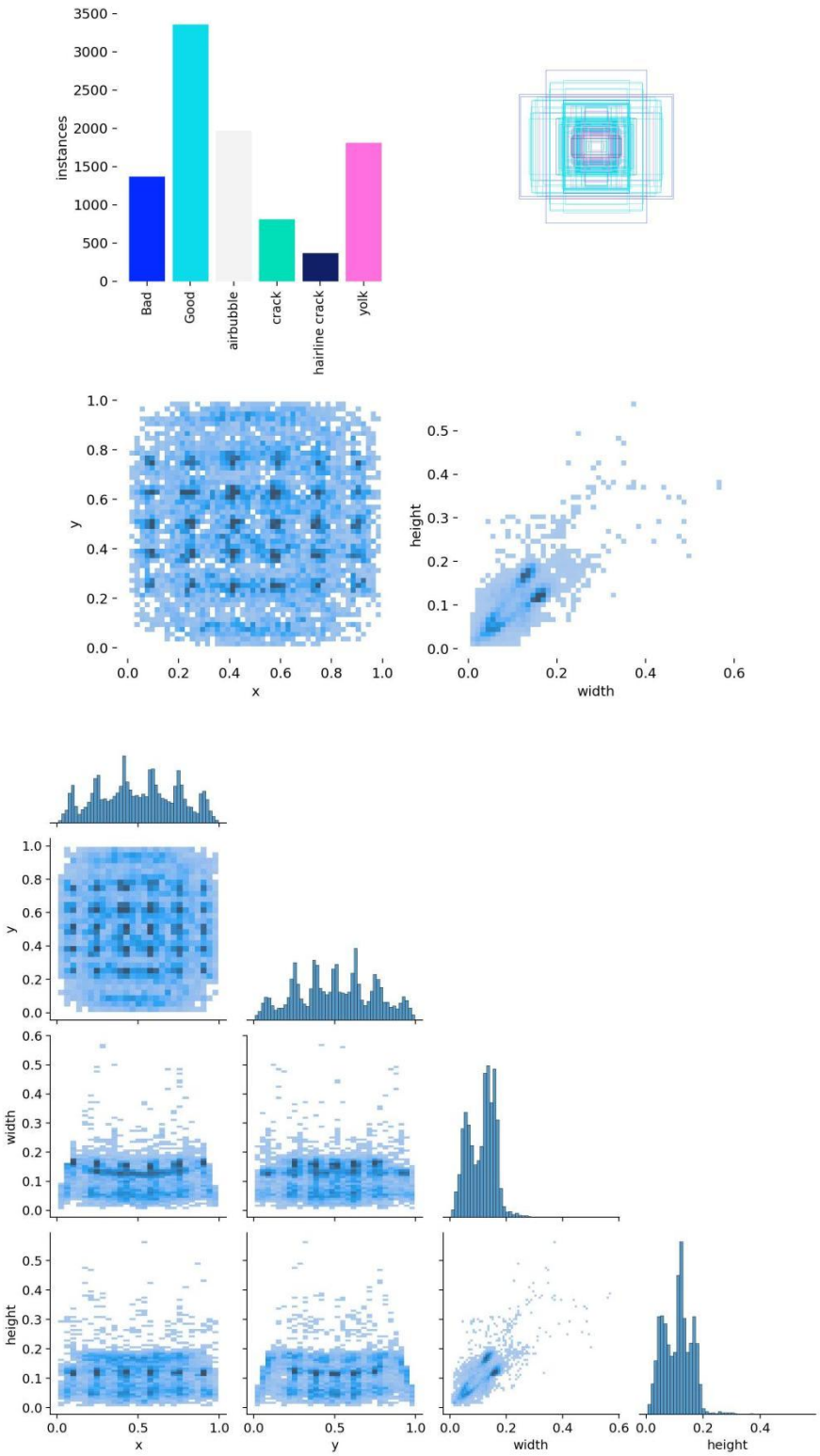
Appendix E

Labels Correlogram Visualization (Baseline Model)



Appendix F

Labels Correlogram Visualization (Enhanced Model)



Appendix G

YOLOv8 Training Source Code

```
!nvidia-smi
```

Show hidden output

```
import os
HOME = os.getcwd()
print(HOME)
```

```
/content
```

```
# Pip install method (recommended)

!pip install ultralytics==8.2.103 -q

from IPython import display
display.clear_output()
# prevent ultralytics from tracking your activity
!yolo settings sync=False
import ultralytics
ultralytics.checks()
```

```
Ultralytics YOLOv8.2.103 🚀 Python-3.12.12 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Setup complete ✅ (2 CPUs, 12.7 GB RAM, 40.0/112.6 GB disk)
```

```
from ultralytics import YOLO

from IPython.display import display, Image
```

```
!mkdir -p {HOME}/datasets
%cd {HOME}/datasets

!pip install roboflow==1.1.48 --quiet

from roboflow import Roboflow
rf = Roboflow(api_key="AzNtVJxGpoEnGgp2IsIu")
project = rf.workspace("training-spwhz").project("duck-egg-external-and-internal")
version = project.version(24)
dataset = version.download("yolov8")
```

Show hidden output

```
%cd {HOME}

!yolo task=segment mode=train model=yolov8l-seg.pt data={dataset.location}/data.yaml epochs=50 imgsz=640
```

Show hidden output

```
%cd {HOME}
Image(filename=f'{HOME}/runs/detect/train/val_batch0_pred.jpg', width=600)
```

Show hidden output

Appendix H

Yolov8 Training Source Code with Real-ESRGAN



```
Commands + Code + Text Run all Connect

import os
import cv2
import numpy as np
from pathlib import Path
from ultralytics import YOLO
import torch


try:
    from realesrgan import RealESRGAN
except Exception:
    RealESRGAN = None


YOLO_WEIGHTS = "duckegganalysis.pt"
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
SCALE = 4
CONF_THRESH = 0.35
SAVE_DIR = Path("out")
SAVE_DIR.mkdir(exist_ok=True)


yolo = YOLO(YOLO_WEIGHTS)
print("YOLO device:", DEVICE)


esrgan = None
if RealESRGAN is not None:

    esrgan = RealESRGAN(device=DEVICE, scale=SCALE)

else:
    print("Warning: RealESRGAN import failed. Adjust import for your installed package.")

def upscale_crop(img_crop):
    """Return upscaled crop (numpy BGR)."""
    if esrgan is None:

        h, w = img_crop.shape[:2]
        return cv2.resize(img_crop, (w * SCALE, h * SCALE), interpolation=cv2.INTER_CUBIC)

    if hasattr(esrgan, "predict") or hasattr(esrgan, "__call__"):

        try:

            rgb = cv2.cvtColor(img_crop, cv2.COLOR_BGR2RGB)
            out = esrgan.enhance(rgb)

            if isinstance(out, np.ndarray):
                out_bgr = cv2.cvtColor(out, cv2.COLOR_RGB2BGR)
                return out_bgr
        except Exception:
            pass

    h, w = img_crop.shape[:2]
    return cv2.resize(img_crop, (w * SCALE, h * SCALE), interpolation=cv2.INTER_CUBIC)

def pad_box(x1, y1, x2, y2, pad_px, width, height):
    return (max(0, int(x1 - pad_px)), max(0, int(y1 - pad_px)),
            min(width, int(x2 + pad_px)), min(height, int(y2 + pad_px)))

def process_image(img_path):
    img = cv2.imread(str(img_path))
    h, w = img.shape[:2]

    results = yolo.predict(source=str(img_path), imgsz=640, conf=CONF_THRESH, device=DEVICE)

    res = results[0]
    boxes = res.boxes
    count = 0
    for box in boxes:
        conf = float(box.conf[0]) if box.conf is not None else 1.0
        if conf < CONF_THRESH:
            continue
        x1, y1, x2, y2 = map(int, box.xyxy[0].tolist())

        pad = int(0.1 * max(x2 - x1, y2 - y1))
        px1, py1, px2, py2 = pad_box(x1, y1, x2, y2, pad, w, h)

        crop = img[py1:py2, px1:px2].copy()
        up = upscale_crop(crop)


        fname = SAVE_DIR / f"{img_path.stem}_det{count}_up.png"
        cv2.imwrite(str(fname), up)

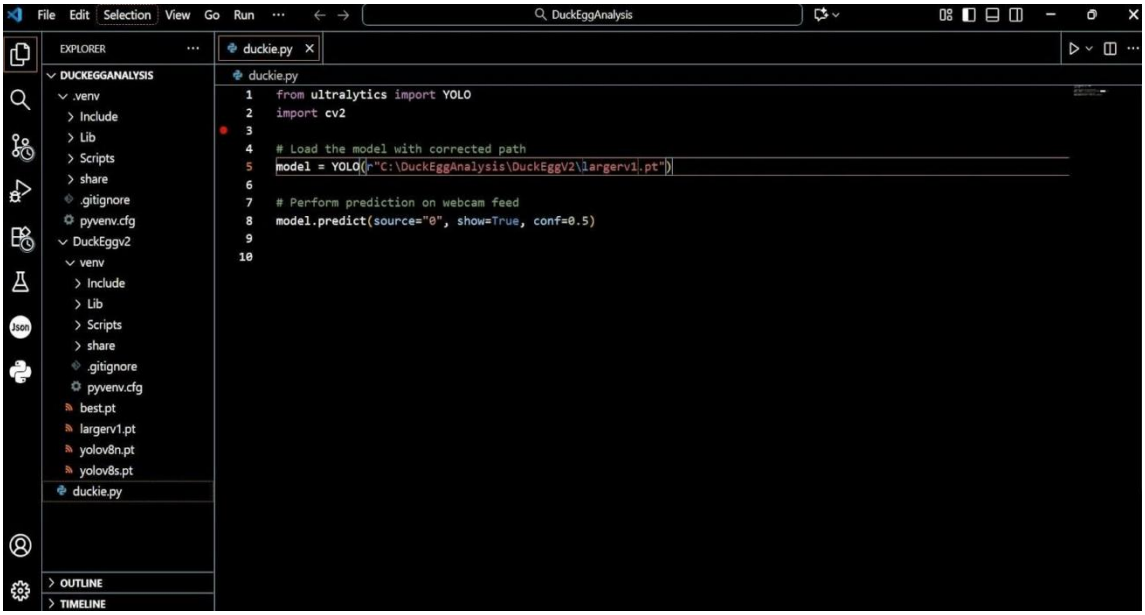

        count += 1

    print(f"Processed {img_path} -> saved {count} upscaled crops")

if __name__ == "__main__":
    import glob
    imgs = list(Path("images").glob("*.jpg")) + list(Path("images").glob("*.png"))
    for p in imgs:
        process_image(p)
```

Appendix I

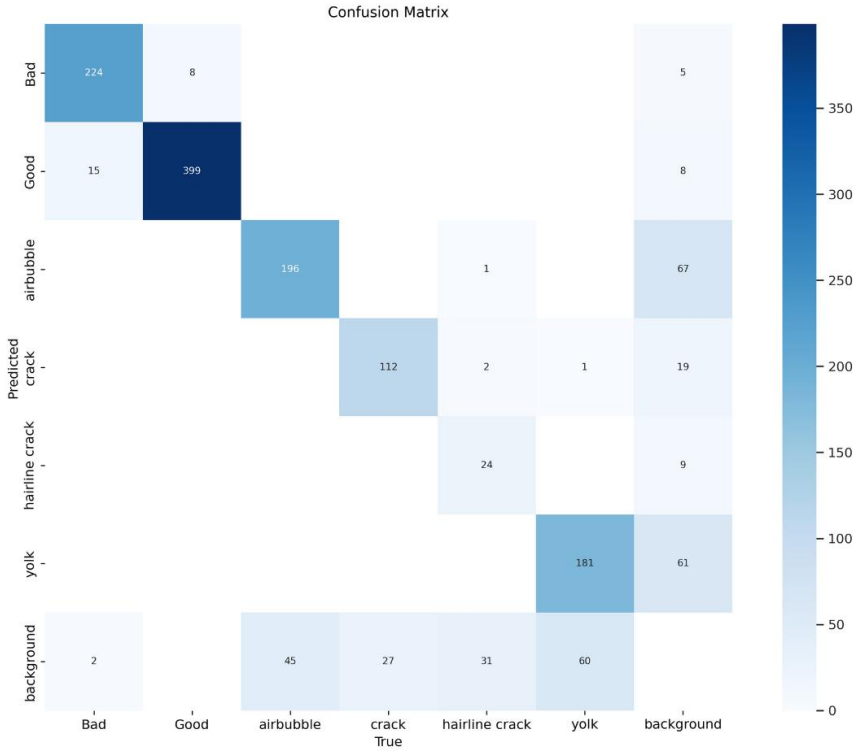
Realtime Detection Source Code of Baseline Model and Enhanced Model



Appendix J

Confusion Matrix and Evaluation Metrics

(Baseline Model)



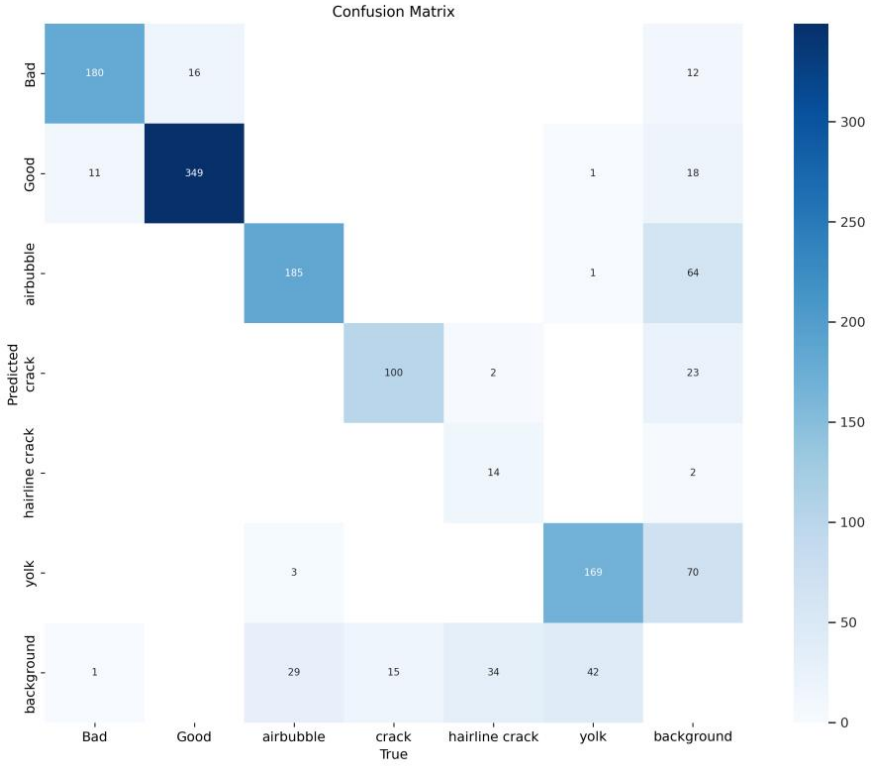

```
50 epochs completed in 0.413 hours.
Optimizer stripped from runs/detect/train/weights/last.pt, 87.7MB
Optimizer stripped from runs/detect/train/weights/best.pt, 87.7MB

Validating runs/detect/train/weights/best.pt...
Ultralytics YOLOv8.2.103 🚀 Python-3.12.12 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 268 layers, 43,611,234 parameters, 0 gradients, 164.8 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 1/1 [00:00<00:00,  1.26it/s]
                   all         27       1328      0.839      0.783      0.824      0.569
                   Bad         20        241      0.961      0.946      0.977      0.937
                  Good         22        407      0.953      0.988      0.985      0.971
            airbubble         22        241      0.747      0.801        0.8       0.42
                 crack         23        139      0.829      0.784      0.829       0.38
        hairline crack         23         58      0.746      0.404      0.503      0.211
                  yolk         22        242      0.799      0.773       0.85      0.493
Speed: 0.2ms preprocess, 17.6ms inference, 0.0ms loss, 1.1ms postprocess per image
Results saved to runs/detect/train
```

Appendix K

Confusion Matrix and Evaluation Metrics

(Enhanced Model)

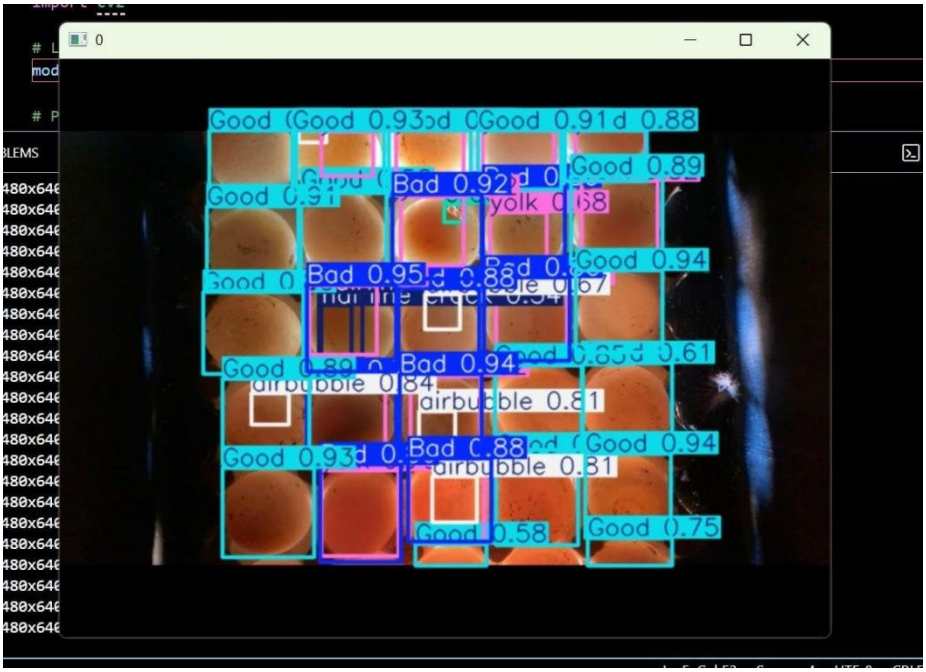



```
50 epochs completed in 0.291 hours.
Optimizer stripped from runs/detect/train/weights/last.pt, 87.7MB
Optimizer stripped from runs/detect/train/weights/best.pt, 87.7MB

Validating runs/detect/train/weights/best.pt...
Ultralytics YOLOv8.2.103 🚀 Python-3.12.12 torch-2.9.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 268 layers, 43,611,234 parameters, 0 gradients, 164.8 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% 1/1 [00:00<00:00,  1.47it/s]
                   all         24       1152      0.823      0.791      0.811      0.575
                   Bad         17        192      0.939      0.956      0.974      0.928
                  Good         19        365      0.952      0.984      0.982       0.97
            airbubble         19        217      0.744      0.816      0.806       0.43
                 crack         20        115      0.826      0.824      0.807      0.423
        hairline crack         20         50      0.706       0.34      0.429      0.194
                  yolk         19        213      0.769      0.826      0.867      0.505
Speed: 0.2ms preprocess, 17.4ms inference, 0.0ms loss, 1.2ms postprocess per image
Results saved to runs/detect/train
```

Appendix L

Real Time Detection Output Result (Baseline Model)



Appendix M

Real Time Detection Output Result (Enhanced Model)

